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Technical Specification

O-RAN Work Group 10 (OAM for O-RAN)

Topology Exposure & Inventory Common Information Models and Interface Specification - Stage 2

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Contents

Foreword

This Technical Specification (TS) has been produced by WG10 of the O-RAN ALLIANCE.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN ALLIANCE modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

version xx.yy.zz

where:

xx: the first digit-group is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have xx=01). Always 2 digits with leading zero if needed.

yy: the second digit-group is incremented when editorial only changes have been incorporated in the document. Always 2 digits with leading zero if needed.

zz: the third digit-group included only in working versions of the document indicating incremental changes during the editing process. External versions never include the third digit-group. Always 2 digits with leading zero if needed.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

Executive summary

The present document specifies the TE&IV Information Models suited to realize the use cases of TE&IV Service Consumer as specified in [1].

Introduction

The O-RAN TE&IV Information Models provide the logical representation of O-RAN specific TE&IV resources. This includes:

- Topology Entities intended to convey classes that realizes TE&IV resources.
- Topology Relationships intended to realize relationship between Topology Entities.
- Enumerations which provide a predefined list of choices.
- Data Types which provide structure to the elements of TE&IV resources.
- Notifications which is the information conveyed in a message from TE&IV Service Producer to TE&IV Service Consumer.

1 Scope

This document specifies the Information Model used to support TE&IV services within the SMO. The TE&IV Information Models described in this document are structured using the concept of Namespaces. In addition, relevant sections in this document include information imported from existing standards and industry work that serve as a basis for O-RAN TE&IV.

The Annex section of this document describes multiple views of the TE&IV Information Models that supports TE&IV Service Consumer use cases [1] within the SMO.

The TE&IV Information Models specified in this present document is part of a “modeling continuum” that aims to establish and evolve an O-RAN TE&IV Information Model from which O-RAN TE&IV Data Models may be generated manually or with a set of tools. However, the O-RAN TE&IV Data Models is out of scope of this present document and will be specified in a different specification.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] O-RAN.WG10.TE&IV-UCR.0: "Topology Exposure and Inventory Management Services Use Cases and Requirement Specification"

[2] 3GPP TS 32.156: Telecommunication management; Fixed Mobile Convergence (FMC) Model repertoire, version 17.5.0, December 2023

[3] OMG formal/2017-12-05: OMG® Unified Modeling Language® (OMG UML®), version 2.5.1

[4] 3GPP TS 32.160: Management and orchestration; Management service template, version 17.9.0, September 2023

[5] 3GPP TS 28.541: 3rd Generation Partnership Project; Technical Specification Group Services and System Aspects; Management and orchestration; 5G Network Resource Model (NRM), version 17.13.0, December 2023

[6] O-RAN.WG1.TS.OAD: " O-RAN Architecture Description" ("OAD")

[7] O-RAN.WG6.TS.O2-GA&P: "O-RAN O2 Interface General Aspects and Principles"

[8] O-RAN.WG6.O2DMS-INTERFACE-ETSI-NFV-PROFILE: "O2dms Interface Specification: Profile based on ETSI NFV Protocol and Data Models"

[9] O-RAN.WG6.O2DMS-INTERFACE-K8S-PROFILE: "O2dms Interface Specification: Kubernetes Native API Profile for Containerized NFs"

[10] O-RAN.WG6.ORCH-USE-CASES: "Cloudification and Orchestration Use Cases and Requirements for O-RAN Virtualized RAN"

[11] O-RAN.WG10.TS.OAM-Architecture: "O-RAN Operations and Maintenance Architecture"

[12] O-RAN.WG3.TS.E2AP: "E2 Application Protocol" ("E2AP")

[13] 3GPP TS 28.658: 3rd Generation Partnership Project; Technical Specification Group Services and System Aspects; Telecommunication management; Evolved Universal Terrestrial Radio Access Network (E-UTRAN) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)

[14] IETF RFC 8141: "Uniform Resource Names (URNs)"

[15] O-RAN.WG4.TS.MP.0: "O-RAN Management Plane Specification"

[16] 3GPP TS 28.622: Generic Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)

[17] O-RAN.WG10.TS.TE&IV-API.0: "Topology Exposure & Inventory Application Protocols Specification - Stage 3" ("TE&IV API")

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

[i.1] 3GPP TR 21.905: "3rd Generation Partnership Project; Technical Specification Group Services and System Aspects; Vocabulary for 3GPP Specifications"

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in [i.1] and the following apply:

Information Model: a representation of concepts and the relationships, constraints, rules, and operations to specify data semantics for a chosen domain of discourse, it specifies relations between objects, can provide sharable, stable, and organized structure of information requirements or knowledge for the domain context.

Namespace: a logical grouping of named elements specified, to prevent name collisions across named elements imported and associated from other specifications to support the independent lifecycle of its representation in the TE&IV Information Model.

Topology Entity: is a modelling construct that represents TE&IV resources useful to a Topology and Inventory use case [1].

Topology Relationship: is a modelling construct that represents the relationship between Topology Entities useful to a Topology and Inventory use case [1].

Domain: a logical grouping of Topology Entities and/or Topology Relationships.

3.2 Symbols

Void

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

TEC Topology Entity Class
TRC Topology Relationship Class

4 TE&IV Information Models

4.1 Introduction

The TE&IV Information Models described in this specification exposes the Topology Entities and Topology Relationships of the TE&IV resources as defined in [1].

4.2 TE&IV Information Modeling Guidelines

4.2.1 Modeling approach, Unified Modeling Language (UML)

The TE&IV Information Models shall use the Unified Modeling Language™ (UML®) version 2.5.1 [3] from the Object Management Group (OMG).

UML provides a rich set of concepts, graphical notations, and model elements to model distributive systems. 3GPP TS 32.156 [2] provides the necessary and sufficient set of UML notations and model elements, including the ones built by the UML extension mechanism <<stereotype>> to model network management systems and their managed nodes. These conventions are applied to the TE&IV Information Model.

The TE&IV Information Model also defines UML notation, model elements and <<stereotypes>> which shall be used to model behaviours of the Topology Entity and Topology Relationship.

The TE&IV Information Model uses embedded PlantUML for diagraming and the relationships are modeled using UML Class diagram conventions.

O-RAN TE&IV Information Models shall also follow methodology documented in 3GPP TS 32.160 [4] Clause 5.2.

4.2.1.1 Model elements and notation

4.2.1.1.1 General

The UML graphical notation in this document is only used to represent the Topology Entity and Topology Relationship model elements.

UML properties as defined in [3] clause 9.5 are referred in this document and shall be used only for TE&IV Information Model.

4.2.1.1.2 Basic model elements

UML [3] has defined a number of basic model elements. This subclause lists the subset selected for use in TE&IV Information Model.

The characteristics defined in UML [3] clause 11.4 is applicable to TE&IV Information Model. However, in this specification, classes may or may not have attributes and the graphical notation of a class may show a suppressed attribute (middle) compartment even if the class has attributes, as shown in figure below. The operation (bottom) compartment is also suppressed.

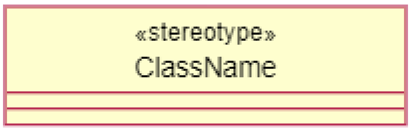

Figure 4.2.1.1.2-1: Basic model elements

4.2.1.1.3 Attribute

4.2.1.1.3.1 Description

In UML [3], an attribute is a typed element of a class that is represented by a property. The properties listed in this section represents the attributes of a Topology Entity and Topology Relationship class.

Table 4.2.1.1.3.1-1 below lists the adaptations of the attribute properties and its values applicable for TE&IV Information Model when re-using the attribute properties specified in 3GPP TS 32.156 [2] clause 5.2.1.

Table 4.2.1.1.3.1-1: TE&IV Information Model valid attribute values

Property name	TE&IV Adaptations
type	This property and its description, legal values shall be supported
allowedValues	This property and its description, legal values shall be supported
defaultValue	This property and its description, legal values shall be supported
multiplicity	This property and its description, legal values shall be supported
isOrdered	This property and its description, legal values shall be supported
isUnique	This property and its description, legal values shall be supported
isNullable	Not required to be supported
passedById	This property and its description, legal values shall be supported
lifecycleStatus	Not required to be supported
isInvariant	This property and its description, legal values shall be supported
isWritable	This property and its description, legal values shall be supported. However, A "isWritable: True" property might be restricted by access control.
isReadable	This property and its description, legal values shall be supported
isNotifyable	This property and its description, legal values shall be supported
supportQualifier	This property and its description, legal values shall be supported

4.2.1.1.3.2 Example

This example shows three attributes, i.e., `x`, `y`, and `z`, listed in the attribute (the second) compartment of the class. However, as mentioned in 4.2.1.1.2, the attribute compartment (middle) can be suppressed.

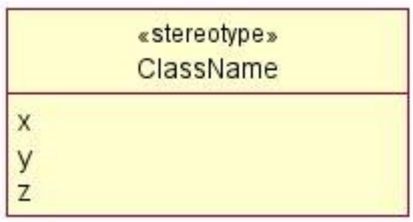

Figure 4.2.1.1.3.2-1: Attribute notation

4.2.1.1.3.3 Name style

As defined in clause 5.2.1.3 of 3GPP TS 32.156 [2].

4.2.1.1.4 Association Relationship

4.2.1.1.4.1 Description

An association relationship class is a modelling construct that represents potentially complex series of associations that may cross domain boundaries. These classes are identifiable, searchable, have attributes and a record of the entities (and potentially configuration/associations) that make up their source.

Association relationship classes are appropriate for use to show a relationship between different Topology Entities [see clause 4.2.1.1.2] through the Topology Relationship class [see clause 4.2.1.1.3]. Each association relationship class has aSide and bSide relationships, multiplicity, and navigability as shown in the example clause 4.2.1.1.4.2.

For a Topology Relationship, the following rules apply:

1. The Topology Relationship class name shall have the form "<X>_{REL}_<Y>", where <X> represents the TEC at the aSide of the relationship, <Y> represents the TEC at the bSide of the relationship, and {REL} is a transient verb representing the relationship between <X> and <Y>. e.g., NFDeployment_serves_ODUFunction.
2. The aSide is considered the originating side of the relationship. The bSide is considered the terminating side of the relationship. The order of aSide and bSide is of importance and shall not be changed once defined.
3. The Topology Relationship is a bi-directional relationship which indicates the navigability from either the aSide or bSide of the associated Topology Entity.

4.2.1.1.4.2 Example

For TE&IV Information model, the example (Figure 4.2.1.1.4.2-1) shows a bi-directional association. In this example, both Topology Entity classes (AbcFunction and XyzFunction) are related through a separate relationship class named Abc_REL_Xyz. By default, the aSide of the relationship is considered originating when relating its associated Topology Entity classes.

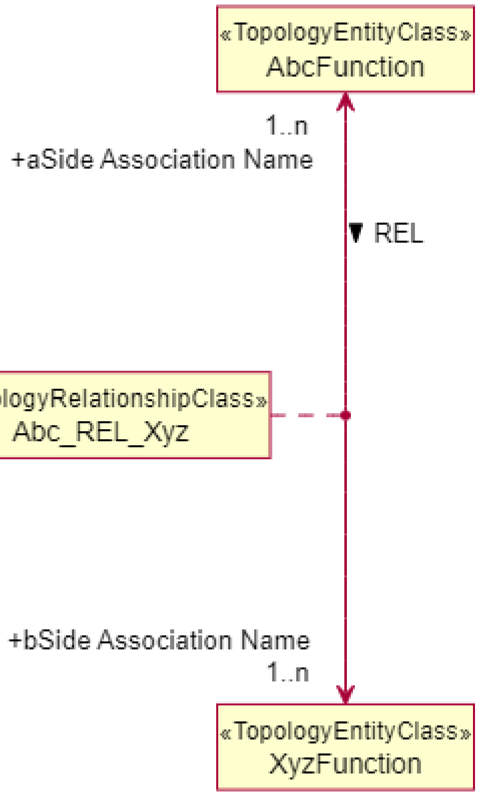

Figure 4.2.1.1.4.2-1: Association Relationship notation

However, the figure 4.2.1.1.4.2-1 can also be represented in a simpler form with only the relationship class name without mentioning association names, as shown in the below figure 4.2.1.1.4.2-2.

Figure 4.2.1.1.4.2-2: Simple Association Relationship notation

4.2.1.1.4.3 Name style

The name shall use the same style as used in << TopologyRelationshipClass >> clause 4.2.1.2.3.3.

4.2.1.1.5 Abstract class

4.2.1.1.5.1 Description

An abstract class is a generalized representation of Topology Entity and Topology Relationship classes. An abstract class cannot be instantiated.

This modelled element has the same properties as class. See 4.2.1.1.2.

4.2.1.1.5.2 Example

This example (Figure 4.2.1.1.5-1) shows that *AbcFunction_* of stereotype TopologyEntityClass is an abstract class. It is a generalization of both AbcFunction1 and AbcFunction2.

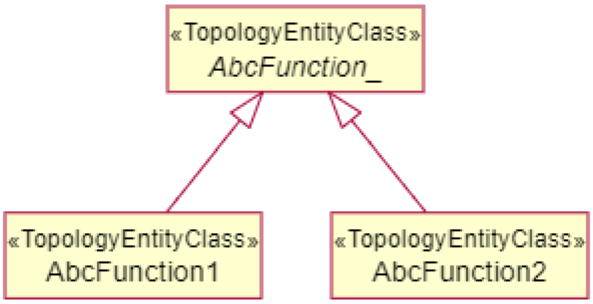

Figure 4.2.1.1.5-1: Abstract class notation for Topology Entity

This example (Figure 4.2.1.1.5-2) shows that `XyzFunction_` of stereotype TopologyRelationshipClass is an abstract class. It is a generalization of both XyzFunction1 and XyzFunction2.

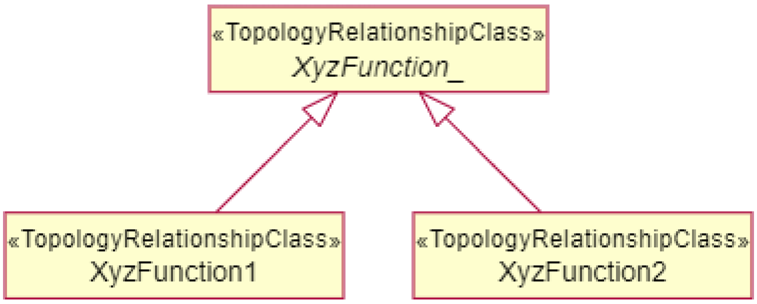

Figure 4.2.1.1.5-2: Abstract class notation for Topology Relationship

4.2.1.1.5.3 Name style

The name shall use the same style as used in TopologyEntityClass and TopologyRelationshipClass as specified in clauses 4.2.1.2.2.3 and 4.2.1.2.3.3. The name shall be in italics and its last character shall be an underscore.

4.2.1.2 Stereotype

4.2.1.2.1 Description

Subclause 4.2.1 listed the UML defined basic model elements. UML defined a stereotype concept allowing the specification of simple or complex user-defined model elements.

This subclause lists all allowable stereotypes for TE&IV Information Model.

The names of stereotypes shall be chosen such that they do not clash with any stereotype defined in O-RAN.

The characteristics defined in subclause 5.3.0 of 3GPP TS 32.156 [2] is applicable to TE&IV Information Model elements i.e, as per the clause 5.3.0 of 3GPP TS 32.156 [2] “For each stereotype model element listed, there are three parts. The first part contains its description. The second part contains its graphical notation examples, and the third part contains the rule, if any, recommended for labelling or naming it”.

4.2.1.2.2 <<TopologyEntityClass>>

4.2.1.2.2.1 Description

The << TopologyEntityClass >> is identical to UML *class* [3] clause 11.4.3.1 which specifies a Topology Entity Resource including properties, attributes and allows inheritance but it does not include/define methods or operations.

4.2.1.2.2.2 Example

This example (figure 4.2.1.2.2.2-1) shows a class `AbcFunction` << TopologyEntityClass >>.

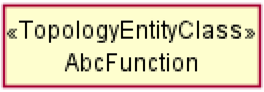

Figure 4.2.1.2.2.2-1: << TopologyEntityClass >> notation

This example (figure 4.2.1.2.2.2-2) shows an abstract class *AbcFunction_* << TopologyEntityClass >>.

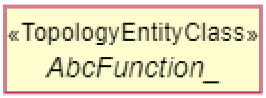

Figure 4.2.1.2.2.2-2: Abstract class << TopologyEntityClass >> notation

4.2.1.2.2.3 Name style

For << TopologyEntityClass >> name, use the same style as defined in clause 5.3.2.3 of 3GPP TS 32.156 [2].

4.2.1.2.3 <<TopologyRelationshipClass>>

4.2.1.2.3.1 Description

The <<TopologyRelationshipClass>> is identical to UML *class* [3] clause 11.4.3.1 which specifies a Topology Relationship between different Topology Entities including properties, attributes and allows multiple inheritance but it does not include/define methods or operations.

4.2.1.2.3.2 Example

This example shows a class `AbcXyzRelation` <<TopologyRelationshipClass>>.

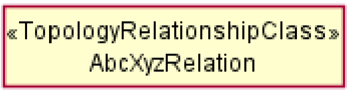

Figure 4.2.1.2.3.2-1: <<TopologyRelationshipClass>> notation

This example shows an abstract class *AbcXyzRelation_* <<TopologyRelationshipClass>>.

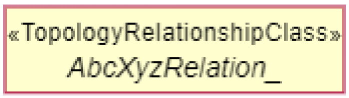

Figure 4.2.1.2.3.2-2: Abstract class <<TopologyRelationshipClass>> notation

4.2.1.2.3.3 Name style

For <<TopologyRelationshipClass>> name, use the same style as defined in clause 5.3.2.3 of 3GPP TS 32.156 [2].

4.2.2 Modeling guidelines

The TE&IV Information Models references concepts from each of the other O-RAN Information Model Namespaces by adding and removing attributes, entities, and relationships to serve the needs of the SMO use cases and services. TE&IV Information Models use multiple separate Namespaces to support independent lifecycle of loosely coupled topology and inventory concepts.

The definitions and mapping involved with the TE&IV resources created within O-RAN TE&IV shall follow the below mentioned guidelines:

- shall be defined in a TE&IV namespace
- shall follow the 3GPP style when defining its identity and attributes [5]
- shall maintain a standard mapping to existing O-RAN definitions within O-RAN namespaces
- if it is a 1:1 mapping with other namespace resource, not limited to O-RAN,
    - the resource name shall be reused exactly
    - the identities used in the TE&IV resources shall encapsulate the type of the instance identifier and the instance identifier of the resource defined in other source namespaces
- if it is a 1:N mapping with other namespace resource, not limited to O-RAN,
    - the resource name shall begin with O-RAN prefix “ORT-”
    - the identities used in the TE&IV resources shall be generated within TE&IV and represented using a TE&IV specified type of the instance identifier and the generated instance identifier of the resource
    - the generated identities to map the TE&IV instance identifier to the source shall be preserved along with the identity of the source.

4.3 TE&IV Information Model Definitions

4.3.1 Namespace ORAN.SMO.TEIV.RAN

4.3.1.1 Namespace overview

This namespace contains the Topology Entities and Topology Relationship in the RAN Logical domain, which represents the functional capability of the deployed RAN that are relevant to realise use cases of TE&IV Service Consumer as specified in [1].

4.3.1.2 Imported associated information

4.3.1.2.1 Imported information entities and local labels

Imported information entities and local labels is not defined in the present version of the document.

4.3.1.2.2 Associated information entities and local labels

Label reference	Local label
TS 28.541 [5], IOC, GNBDUFunction	GNBDUFunction
TS 28.541 [5], IOC, GNBCUCPFunction	GNBCUCPFunction
TS 28.541 [5], IOC, GNBCUUPFunction	GNBCUUPFunction

4.3.1.3 Class diagram

4.3.1.3.1 Relationships

Relationships are not defined in the present version of the document.

4.3.1.3.2 Inheritance

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members
set namespaceSeparator none

namespace ORAN.SMO.TEIV {

abstract class TopologyEntity_
}

namespace ORAN.SMO.TEIV.RAN {

abstract class ORANNetworkFunction_
class ORUFunction
class ODUFunction
class OCUCPFunction
class OCUUPFunction
class NearRTRICFunction

TopologyEntity_ <|-d- ORANNetworkFunction_

ORANNetworkFunction_ <|-- ORUFunction
ORANNetworkFunction_ <|-- ODUFunction
ORANNetworkFunction_ <|-- OCUCPFunction
ORANNetworkFunction_ <|-- OCUUPFunction
ORANNetworkFunction_ <|-- NearRTRICFunction
}
@enduml
```

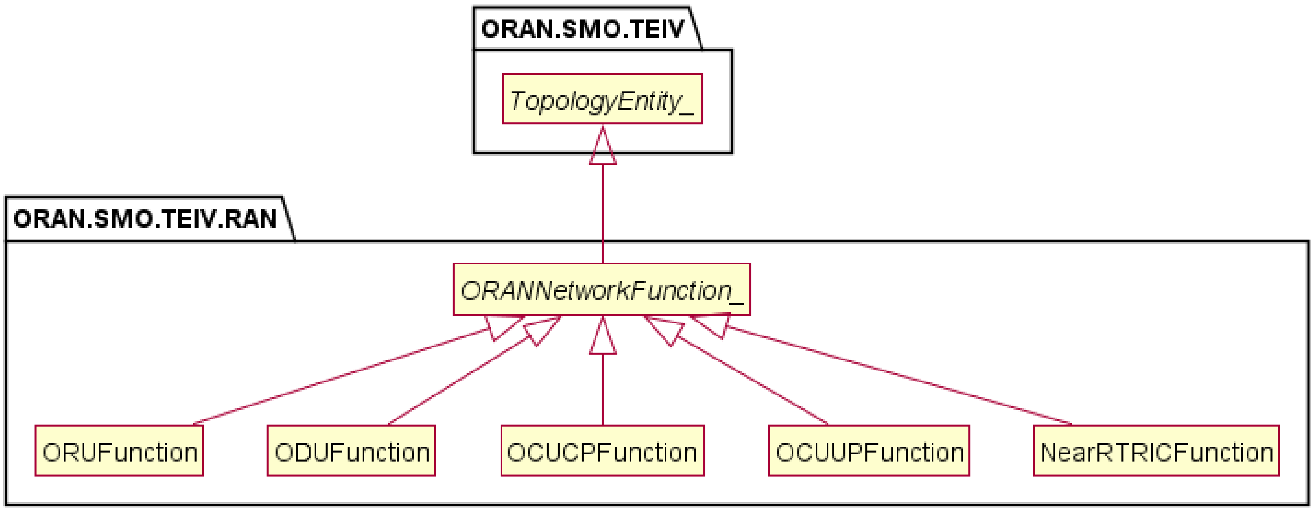

Figure 4.3.1.3.2-1: ORAN SMO TE&IV RAN inheritance view

4.3.1.4 Class definitions

4.3.1.4.1 ORUFunction

4.3.1.4.1.1 Definitions

This class provides the TE&IV resource representation of O-RU O-RAN Network Function using the equivalent concept as defined in clause 5 of O-RAN.WG1.OAD [6].

4.3.1.4.1.2 Attributes

The ORUFunction TEC includes the attributes inherited from *ORANNetworkFunction_* and have the following attributes:

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
oruId	M	T	F	F	T

4.3.1.4.1.3 Attribute constraints

None.

4.3.1.4.1.4 Notifications

There is no notification defined.

4.3.1.4.1.5 State diagram

None.

4.3.1.4.2 NearRTRICFunction

4.3.1.4.2.1 Definitions

This class provides the TE&IV resource representation of Near-RT RIC O-RAN Network Function using the equivalent concept as defined in clause 5 of O-RAN.WG1.OAD [6].

4.3.1.4.2.2 Attributes

The NearRTRICFunction TEC includes the topology attributes inherited from *ORANNetworkFunction_* and have the following attributes:

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
pLMNId	M	T	F	F	T
nearRtRicId	M	T	F	F	T

4.3.1.4.2.3 Attribute constraints

None.

4.3.1.4.2.4 Notifications

There is no notification defined.

4.3.1.4.2.5 State diagram

None.

4.3.1.4.3 OCUCPFunction

4.3.1.4.3.1 Definitions

This class provides the TE&IV resource representation of O-CU-CP Network Function using the equivalent concept as defined in clause 5 of O-RAN TS OAD [6].

NOTE: The management characteristics of the O-CU-CP O-RAN NF is represented by the IOC GNBCUCPFunction as specified in 3GPP NR NRM model, as per 3GPP TS 28.541 [5].

4.3.1.4.3.2 Attributes

The OCUCPFunction TEC includes the attributes inherited from *ORANNetworkFunction_* and have the following attributes:

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
gNBCUName	O	T	F	F	T
gNBId	M	T	F	F	T
gNBIdLength	M	T	F	F	T
pLMNId	M	T	F	F	T

4.3.1.4.3.3 Attribute constraints

None.

4.3.1.4.3.4 Notifications

There is no notification defined.

4.3.1.4.3.5 State diagram

None.

4.3.1.4.4 OCUUPFunction

4.3.1.4.4.1 Definitions

This class provides the TE&IV resource representation of O-CU-UP Network Function using the equivalent concept as defined in clause 5 of O-RAN.WG1.OAD [6].

NOTE: The management characteristics of the O-CU-UP O-RAN NF is represented by the IOC GNBCUUPFunction as specified in 3GPP NR NRM model, as per 3GPP TS 28.541 [5].

4.3.1.4.4.2 Attributes

The OCUUPFunction TEC includes the attributes inherited from *ORANNetworkFunction_* and have the following attributes:

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
gNBId	M	T	F	F	T
gNBIdLength	M	T	F	F	T
pLMNIdList	M	T	F	F	T

4.3.1.4.4.3 Attribute constraints

None.

4.3.1.4.4.4 Notifications

There is no notification defined.

4.3.1.4.4.5 State diagram

None.

4.3.1.4.5 ODUFunction

4.3.1.4.5.1 Definitions

This class provides the TE&IV resource representation of O-DU Network Function using the equivalent concept as defined in clause 5 of O-RAN.WG1.OAD [6].

NOTE: The management characteristics of the O-DU O-RAN NF is represented by the IOC GNBDUFunction as specified in 3GPP NR NRM model, as per 3GPP TS 28.541 [5].

4.3.1.4.5.2 Attributes

The ODUFunction TEC includes the attributes inherited from *ORANNetworkFunction_* and have the following attributes:

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
gNBDUId	M	T	F	F	T
gNBId	M	T	F	F	T
gNBIdLength	M	T	F	F	T

4.3.1.4.5.3 Attribute constraints

None.

4.3.1.4.5.4 Notifications

There is no notification defined.

4.3.1.4.5.5 State diagram

None.

4.3.1.4.6 *ORANNetworkFunction_*

4.3.1.4.6.1 Definitions

This abstract class is provided for sub-classing only and is used to generalize the entity classes in this namespace.

4.3.1.4.6.2 Attributes

The *ORANNetworkFunction_* includes the attributes inherited from *TopologyEntity_* in the ORAN.SMO.TEIV namespace.

4.3.1.4.6.3 Attribute constraints

None.

4.3.1.4.6.4 Notifications

There is no notification defined.

4.3.1.4.6.5 State diagram

None.

4.3.1.5 Attribute definitions

4.3.1.5.1 Attribute properties

The following table defines the properties of attributes specified in the present document.

Table 4.3.1.5.1-1: Attribute properties

Attribute Name	Documentation and Allowed Values	Properties	Source Domain
gNBCUName	Name of gNB-CU, see subclause 4.4.1 of 3GPP TS 28.541 [5]. allowedValues: Not applicable	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None	O1
gNBDUId	Unique identifier for the gNB-DU at least within a gNB-CU, see subclause 4.4.1 of 3GPP TS 28.541 [5]. allowedValues: Not applicable	type: Integer multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None	O1
gNBId	Identity of gNB within a PLMN, see subclause 4.4.1 of 3GPP TS 28.541 [5]. allowedValues: 0..4294967295	type: Integer multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None	O1
gNBIdLength	Length of gNBId bit string representation, see subclause 4.4.1 of 3GPP TS 28.541 [5]. allowedValues: 22 .. 32	type: Integer multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None	O1
nearRTRICID	The identifier of Near-RT RIC as defined in O-RAN.WG3.E2AP [12], clause 9.2.4	type: Integer multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None	O1
pLMNId	Identity of the PLMN, see subclause 4.4.1 of 3GPP TS 28.541 [5].	type: PLMNId multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None	O1
oruId	Identity of the O-RU as discovered from the source domain based on M-Plane architecture model as defined in O-RAN TS [15] Note: The oruId is assumed to be available in TE&IV.	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None	O1 (Hierarchical) O1, Open FH M-Plane (Hybrid)
pLMNIdList	List of unique identities for PLMN, see subclause 4.4.1 of 3GPP TS 28.658 [13].	type: PLMNId multiplicity: 1..12 isOrdered: False isUnique: True isWritable: False defaultValue: None	O1

4.3.2 Namespace ORAN.SMO.TEIV.Cloud

4.3.2.1 Namespace overview

This namespace contains the Topology Entities and Topology Relationship in the O-CLOUD domain, which comprises cloud infrastructure and cloud deployment aspects.

4.3.2.2 Imported associated information

4.3.2.2.1 Imported information entities and local labels

Imported information entities and local labels is not defined in the present version of the document.

4.3.2.2.2 Associated information entities and local labels

Associated information entities and local labels is not defined in the present version of the document.

4.3.2.3 Class diagram

4.3.2.3.1 Relationships

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members
set namespaceSeparator none
namespace ORAN.SMO.TEIV.Cloud {

class CloudifiedNF
class NFDeployment
class OCloudNamespace
class NodeCluster
class OCloudSite

note as locationnote
  Has location info.
end note

locationnote .. OCloudSite


CloudifiedNF "1" --> "1..*" NFDeployment: comprizes
NFDeployment " 1..* "--> "1..*" OCloudNamespace : deployedOn
OCloudNamespace " 1..* "--> "1" NodeCluster : deployedOn
NodeCluster "1..*" --> "1..*" OCloudSite : locatedAt
}
@enduml
```


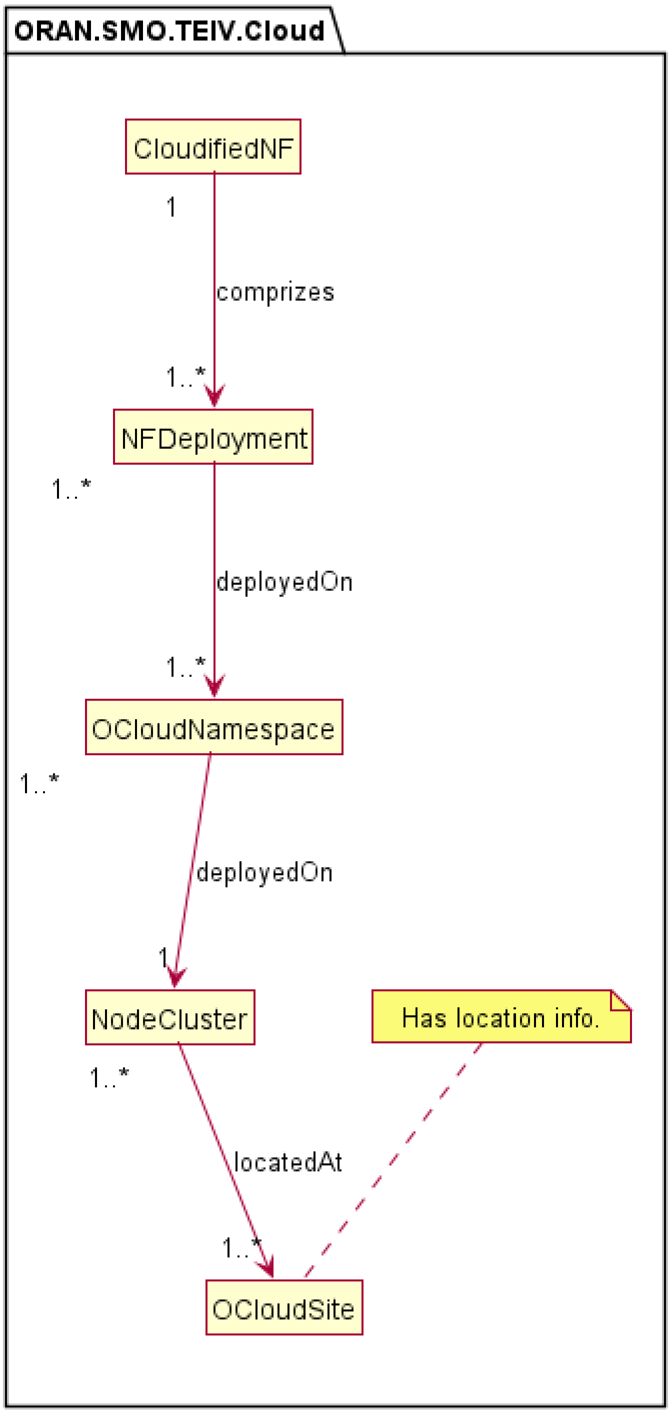

Figure 4.3.2.3.1-1: ORAN SMO TE&IV O-Cloud relationship view

4.3.2.3.2 Inheritance

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members
set namespaceSeparator none
namespace ORAN.SMO.TEIV.Cloud {

class Resource
```



```
class CloudifiedNF
class NFDeployment
class OCloudNamespace
class NodeCluster
class OCloudSite

Resource <|-- CloudifiedNF
Resource <|-- NFDeployment
Resource <|-- OCloudNamespace
Resource <|-- NodeCluster
Resource <|-- OCloudSite
}
@enduml
```

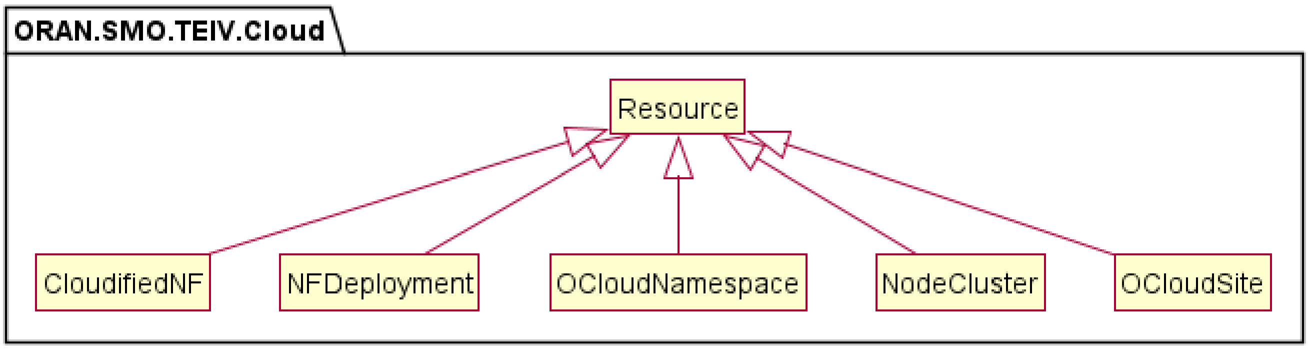

Figure 4.3.2.3.2-1: ORAN SMO TE&IV O-Cloud inheritance view

4.3.2.4 Class definitions

4.3.2.4.1 CloudifiedNF

4.3.2.4.1.1 Definitions

This class provides the TE&IV resource representation of Cloudified NF using the equivalent concept as defined in O-RAN.WG6.O2-GA&P [7].

4.3.2.4.1.2 Attributes

Not specified in the present version of the document.

4.3.2.4.1.3 Attribute constraints

Not specified in the present version of the document.

4.3.2.4.1.4 Notifications

Not specified in the present version of the document.

4.3.2.4.1.5 State diagram

Not specified in the present version of the document.

4.3.2.4.2 NFDeployment

4.3.2.4.2.1 Definitions

This class provides the TE&IV resource representation of NF Deployment using the equivalent concept as defined in O-RAN.WG6.O2-GA&P [7].

4.3.2.4.2.2 Attributes

Not specified in the present version of the document.

4.3.2.4.2.3 Attribute constraints

Not specified in the present version of the document.

4.3.2.4.2.4 Notifications

Not specified in the present version of the document.

4.3.2.4.2.5 State diagram

Not specified in the present version of the document.

4.3.2.4.3 OCloudNamespace

4.3.2.4.3.1 Definitions

This class provides the TE&IV resource representation of OCloud Namespace using the equivalent concept as “containerNamespace” defined in O-RAN.WG6.O2DMS-INTERFACE-ETSI-NFV-PROFILE [8] and as “namespace” in O-RAN.WG6.O2DMS-INTERFACE-K8S-PROFILE [9].

4.3.2.4.3.2 Attributes

Not specified in the present version of the document.

4.3.2.4.3.3 Attribute constraints

Not specified in the present version of the document.

4.3.2.4.3.4 Notifications

Not specified in the present version of the document.

4.3.2.4.3.5 State diagram

Not specified in the present version of the document.

4.3.2.4.4 NodeCluster

4.3.2.4.4.1 Definitions

This class provides the TE&IV resource representation of O-Cloud Node Cluster using the equivalent concept as defined in O-RAN.WG6.O2-GA&P [7].

4.3.2.4.4.2 Attributes

Not specified in the present version of the document.

4.3.2.4.4.3 Attribute constraints

Not specified in the present version of the document.

4.3.2.4.4.4 Notifications

Not specified in the present version of the document.

4.3.2.4.4.5 State diagram

Not specified in the present version of the document.

4.3.2.4.5 OCloudSite

4.3.2.4.5.1 Definitions

This class provides the TE&IV resource representation of O-Cloud Site using the equivalent concept as defined in O-RAN.WG6.O2-GA&P [7].

4.3.2.4.5.2 Attributes

Not specified in the present version of the document.

4.3.2.4.5.3 Attribute constraints

Not specified in the present version of the document.

4.3.2.4.5.4 Notifications

Not specified in the present version of the document.

4.3.2.4.5.5 State diagram

Not specified in the present version of the document.

4.3.3 Namespace ORAN.SMO.TEIV.REL-Cloud-RAN

4.3.3.1 Namespace overview

This namespace contains the relationship between ORAN.SMO.TEIV.RAN to ORAN.SMO.TEIV.Cloud namespaces.

The *ORANNetworkFunction_* abstract class depicted in the figure 4.3.3.3.1-1 is a generalization of the various O-RAN Network Functions and shall be associated from the ORAN.SMO.TEIV.RAN namespace.

The *NFDeployment_REL_NF_* abstract class depicted in the figure 4.3.3.3.1-1 is a generalization of the various relationships between the O-RAN Network Functions and its associated NF Deployments.

The NFDeployment class depicted in the figure 4.3.3.3.1-1 shall be associated from the ORAN.SMO.TEIV.Cloud namespace.

4.3.3.2 Imported associated information

4.3.3.2.1 Imported information entities and local labels

Imported information entities and local labels is not defined in the present version of the document.

4.3.3.2.2 Associated information entities and local labels

Associated information entities and local labels is not defined in the present version of the document.

4.3.3.3 Class diagram

4.3.3.3.1 Relationships

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members
set namespaceSeparator none

namespace ORAN.SMO.TEIV.RAN {
abstract class ORANNetworkFunction_

}
namespace ORAN.SMO.TEIV.REL-Cloud-RAN {
abstract class NFDeployment_REL_NF_

}
namespace ORAN.SMO.TEIV.Cloud {
class NFDeployment

}

NFDeployment  "1..n" <-> "1..n" ORANNetworkFunction_
NFDeployment_REL_NF_ "1" -[hidden]d- NFDeployment
NFDeployment_REL_NF_ -[hidden]d- ORANNetworkFunction_
NFDeployment_REL_NF_ . (NFDeployment, ORANNetworkFunction_)
@enduml
```

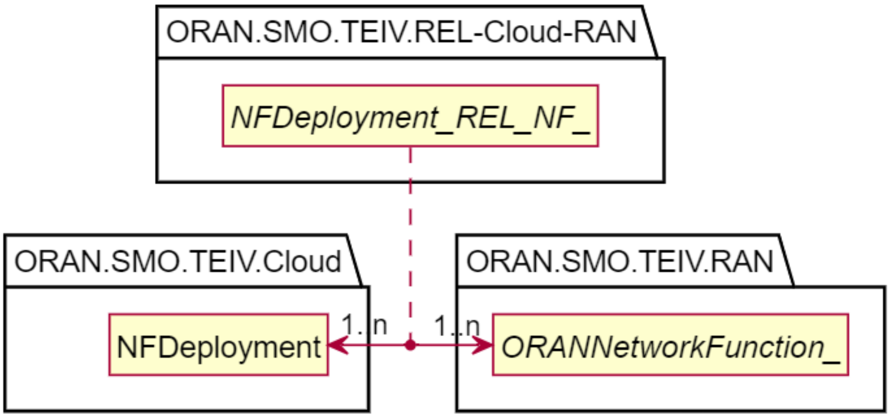

Figure 4.3.3.3.1-1: SMO TE&IV Cloud-RAN relationship model

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members
set namespaceSeparator none

namespace ORAN.SMO.TEIV.RAN {
abstract class ORANNetworkFunction_

}

namespace ORAN.SMO.TEIV.Cloud {
class NFDeployment

}
NFDeployment <-right-> ORANNetworkFunction_:\t<b><i> NFDeployment_REL_NF_\t\t
@enduml
```

Figure 4.3.3.3.1-2: Simplified SMO TE&IV Cloud-RAN relationship model

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members
set namespaceSeparator none

namespace ORAN.SMO.TEIV.RAN {
class ODUFunction{
    }

}
namespace ORAN.SMO.TEIV.REL-Cloud-RAN {
class NFDeployment_serves_ODUFunction{
    }
}
namespace ORAN.SMO.TEIV.Cloud {
class NFDeployment{
    }
```



```
}

NFDeployment  " \n1..n\nserved-oduFunction" <---> "serving-nfDeployment\n1..n\n\n   "
ODUFunction: > serves
NFDeployment_serves_ODUFunction . (NFDeployment, ODUFunction)
@enduml
```

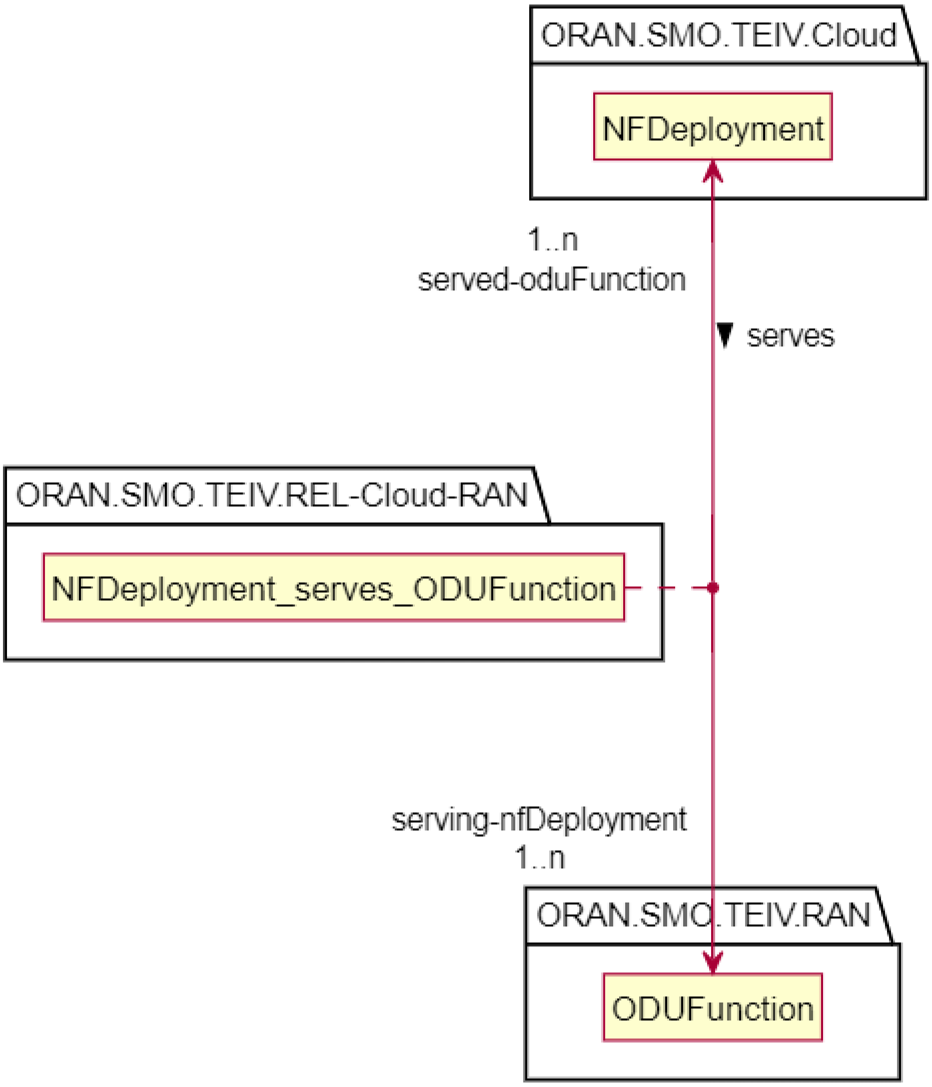

Figure 4.3.3.3.1-3: NFDeployment and ODUFunction relationship model

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members
set namespaceSeparator none

namespace ORAN.SMO.TEIV.RAN {
class OCUCPFunction{
    }

}
namespace ORAN.SMO.TEIV.REL-Cloud-RAN {
```



```
class NFDeployment_serves_OCUCPFunction{
     }
}
namespace ORAN.SMO.TEIV.Cloud {
class NFDeployment{
     }
}

NFDeployment  " \n1..n\nserved-ocucpFunction" <---> "serving-nfDeployment\n1..n\n\n   "
OCUCPFunction: > serves
NFDeployment_serves_OCUCPFunction . (NFDeployment, OCUCPFunction)
@enduml
```

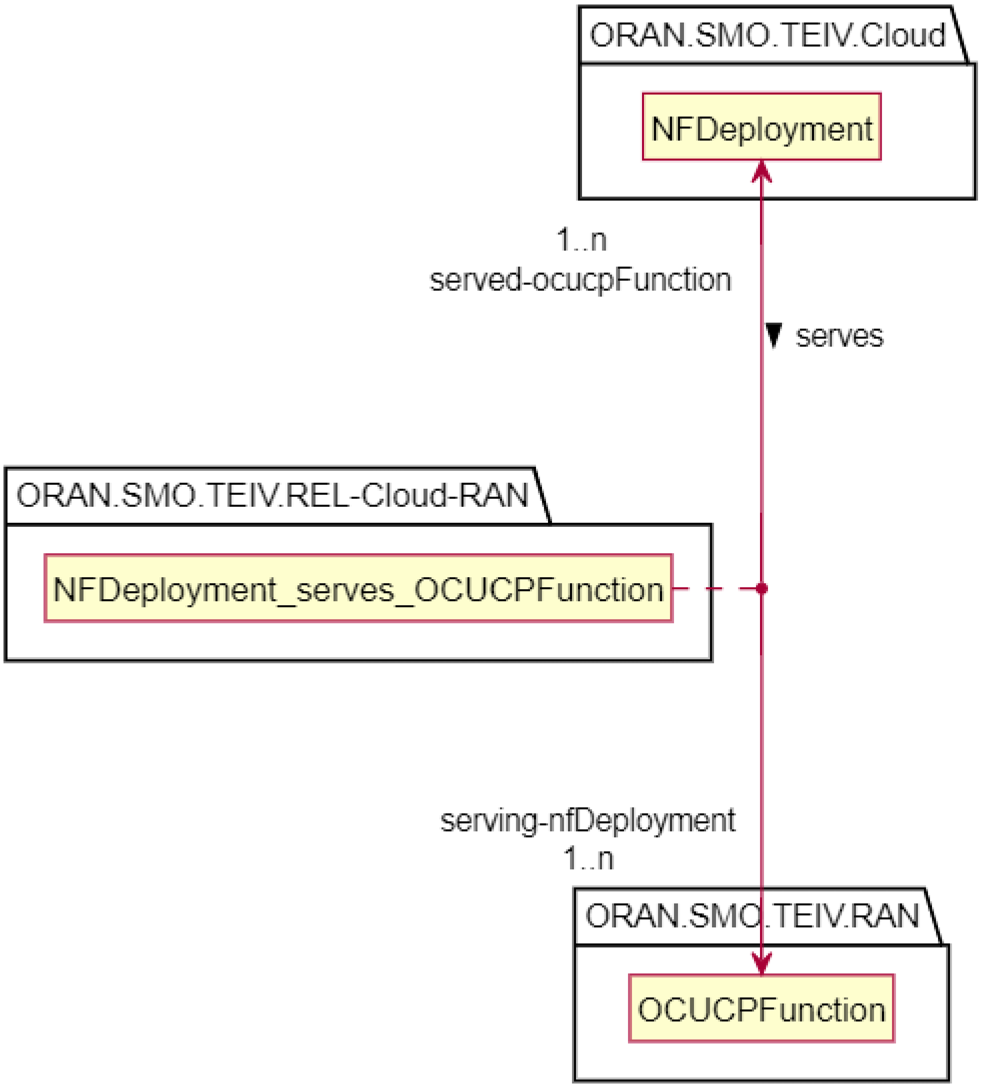

Figure 4.3.3.3.1-4: NFDeployment and OCUCPFunction relationship model

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members
set namespaceSeparator none

namespace ORAN.SMO.TEIV.RAN {
class OCUUPFunction{
     }
```



```
}
namespace ORAN.SMO.TEIV.REL-Cloud-RAN {
class NFDeployment_serves_OCUUPFunction{
    }
}
namespace ORAN.SMO.TEIV.Cloud {
class NFDeployment{
    }
}

NFDeployment  " \n1..n\nserved-ocuupFunction" <---> "serving-nfDeployment\n1..n\n\n   "
OCUUPFunction: > serves
NFDeployment_serves_OCUUPFunction . (NFDeployment, OCUUPFunction)
@enduml
```

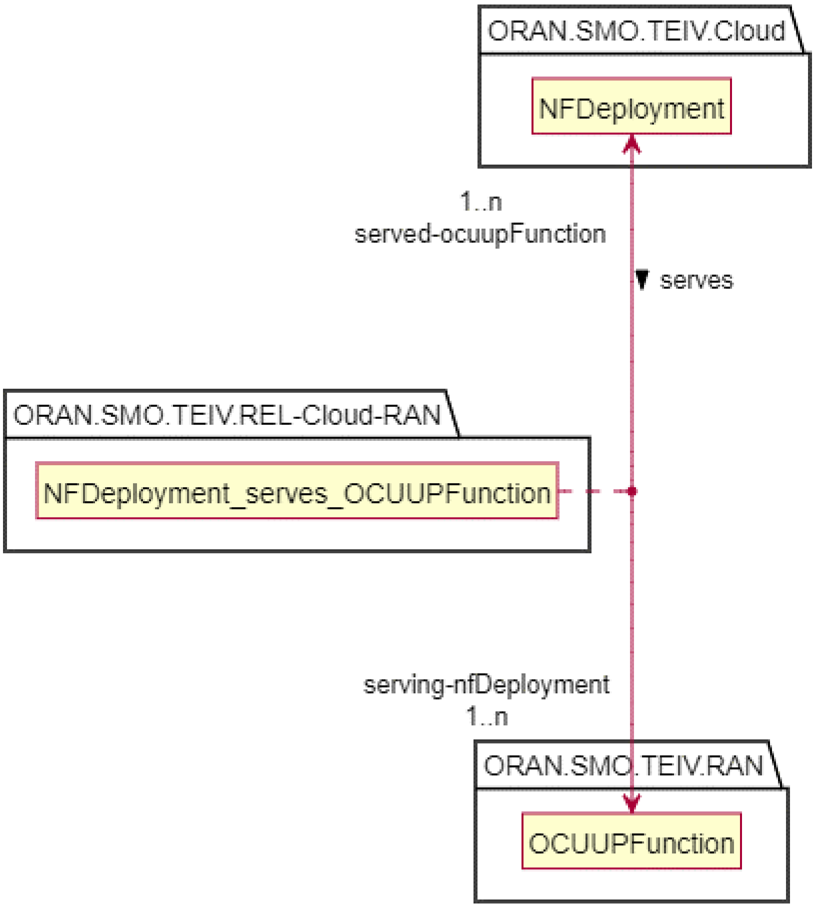

Figure 4.3.3.3.1-5: NFDeployment and OCUUPFunction relationship model

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members
set namespaceSeparator none

namespace ORAN.SMO.TEIV.RAN {
class NearRTRICFunction{
    }

}
namespace ORAN.SMO.TEIV.REL-Cloud-RAN {
class NFDeployment_serves_NearRTRICFunction{
    }
```



```
}
namespace ORAN.SMO.TEIV.Cloud {
class NFDeployment{
    }
}

NFDeployment  " \n1..n\nserved-nearRTRICFunction" <---> "serving-nfDeployment\n1..n\n\n   "
NearRTRICFunction: > serves
NFDeployment_serves_NearRTRICFunction . (NFDeployment, NearRTRICFunction)
@enduml
```

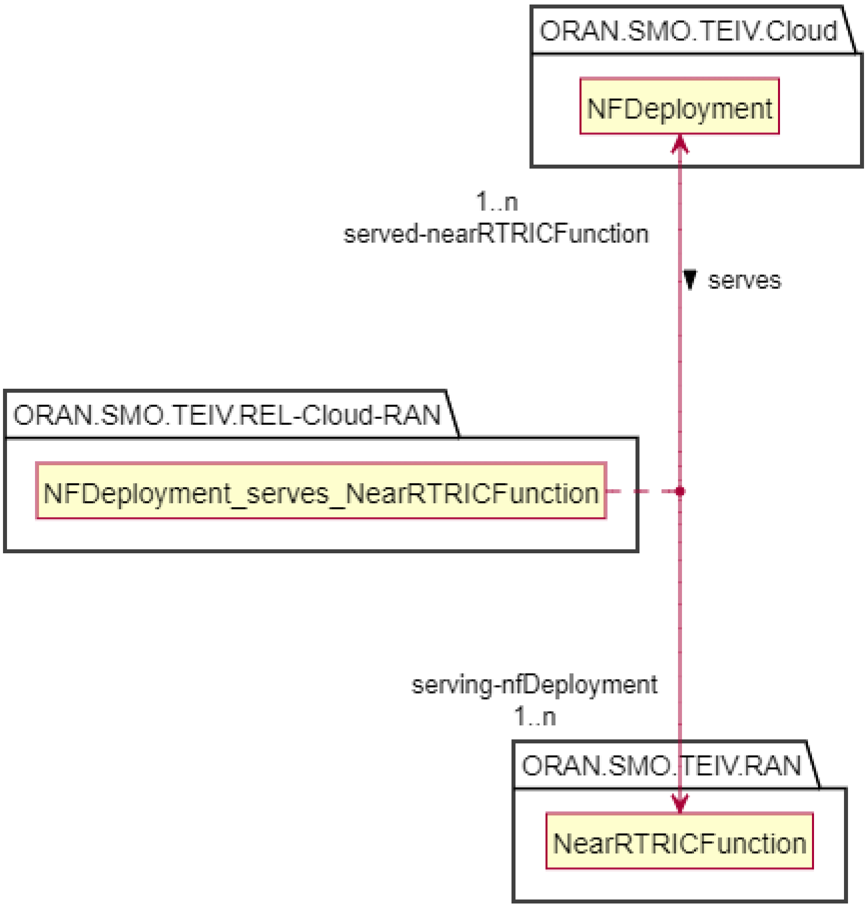

Figure 4.3.3.3.1-6: NFDeployment and NearRTRICFunction relationship model

4.3.3.3.2 Inheritance

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members
set namespaceSeparator none
namespace ORAN.SMO.TEIV {
    abstract class Topology_REL_ {}
}
namespace ORAN.SMO.TEIV.REL-Cloud-RAN {

abstract class NFDeployment_REL_NF_ {}

Topology_REL_ <|-- NFDeployment_REL_NF_
NFDeployment_REL_NF_ <|-- NFDeployment_serves_ODUFunction
NFDeployment_REL_NF_ <|-- NFDeployment_serves_OCUCPFunction
NFDeployment_REL_NF_ <|-- NFDeployment_serves_OCUUPFunction
```



```
NFDeployment_REL_NF_ <|-- NFDeployment_serves_NeartRTRICFunction
}

@enduml
```

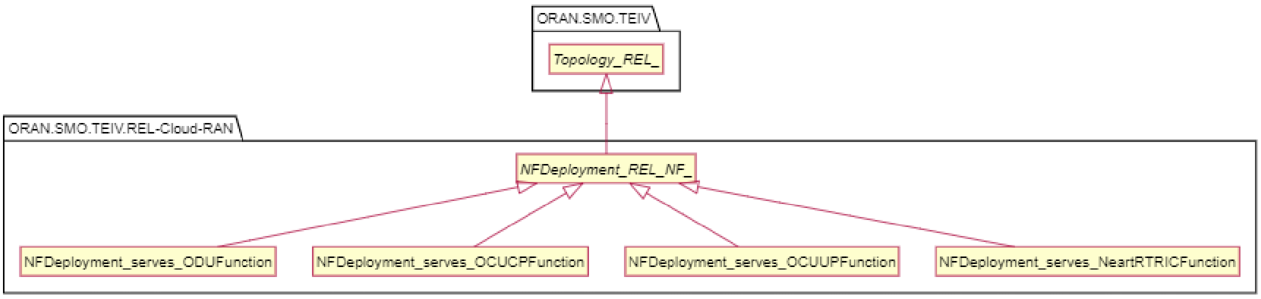

Figure 4.3.3.3.2-1: TE&IV Cloud-RAN Serves Relationship Generalization

4.3.3.4 Class definitions

4.3.3.4.1 NFDeployment_serves_ODUFunction

4.3.3.4.1.1 Definitions

This class provides the Topology Relationship between the NFDeployment and the ODUFunction Topology Entities as shown in figure 4.3.3.3.1-3. This class represents the relationship type of the NF Deployment serving the functionality of the ODU Function.

4.3.3.4.1.2 Attributes

The NFDeployment_serves_ODUFunction TRC includes the attributes inherited from *TopologyRel_* and has the following attributes:

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
served-oduFunction	M	T	F	F	T
serving-nfDeployment	M	T	F	F	T

4.3.3.4.1.3 Attribute constraints

None.

4.3.3.4.1.4 Notifications

None.

4.3.3.4.1.5 State diagram

None.

4.3.3.4.2 NFDeployment_serves_OCUCPFunction

4.3.3.4.2.1 Definitions

This class provides the Topology Relationship between the NFDeployment and the OCUCPFunction Topology Entities as shown in figure 4.3.3.3.1-4. This class represents the relationship type of the NF Deployment serving the functionality of the OCUCP Function.

4.3.3.4.2.2 Attributes

The NFDeployment_serves_OCUCPFunction TRC includes the attributes inherited from *TopologyRel_* and has the following attributes:

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
served-ocucpFunction	M	T	F	F	T
serving-nfDeployment	M	T	F	F	T

4.3.3.4.2.3 Attribute constraints

None.

4.3.3.4.2.4 Notifications

None.

4.3.3.4.2.5 State diagram

None.

4.3.3.4.3 NFDeployment_serves_OCUUPFunction

4.3.3.4.3.1 Definitions

This class provides the Topology Relationship between the NFDeployment and the OCUUPFunction Topology Entities as shown in figure 4.3.3.3.1-5. This class represents the relationship type of the NF Deployment serving the functionality of the OCUUP Function.

4.3.3.4.3.2 Attributes

The NFDeployment_serves_OCUUPFunction TRC includes the attributes inherited from *TopologyRel_* and has the following attributes:

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
served-ocuupFunction	M	T	F	F	T
serving-nfDeployment	M	T	F	F	T

4.3.3.4.3.3 Attribute constraints

None.

4.3.3.4.3.4 Notifications

None.

4.3.3.4.3.5 State diagram

None.

4.3.3.4.4 NFDeployment_serves_NearRTRICFunction

4.3.3.4.4.1 Definitions

This class provides the Topology Relationship between the NFDeployment and the NearRTRICFunction Topology Entities as shown in figure 4.3.3.3.1-6. This class represents the relationship type of the NF Deployment serving the functionality of the Near-RT RIC Function.

4.3.3.4.4.2 Attributes

The NFDeployment_serves_ NearRTRICFunction TRC includes the attributes inherited from *TopologyRel_* and has the following attributes:

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
served-nearRTRICFunction	M	T	F	F	T
serving-nfDeployment	M	T	F	F	T

4.3.3.4.4.3 Attribute constraints

None.

4.3.3.4.4.4 Notifications

None.

4.3.3.4.4.5 State diagram

None.

4.3.3.4.4 *NFDeployment_REL_NF_*

4.3.3.4.4.1 Definitions

This abstract class is provided for sub-classing only and is used to generalize the relationship classes in this namespace.

4.3.3.4.4.2 Attributes

None.

4.3.3.4.4.3 Attribute constraints

None.

4.3.3.4.4.4 Notifications

None.

4.3.3.4.4.5 State diagram

None.

4.3.3.5 Attribute definitions

4.3.3.5.1 Attribute properties

The following table defines the properties of attributes specified in the present document.

Table 4.3.3.5.1-1: Attribute properties

Attribute Name	Documentation and Allowed Values	Properties	Source Domain
serving-nfDeployment	This represents the Topology Entity Id(s) of the NF Deployment instance(s) serving an O-RAN Network Function.	type: URN multiplicity: 1..n isOrdered: False isUnique: True defaultValue: None	TE&IV
served-oduFunction	This represents the Topology Entity Id(s) of the ODU Function instance(s) served by an NF Deployment.	type: URN multiplicity: 1..n isOrdered: False isUnique: True defaultValue: None	TE&IV
served-ocucpFunction	This represents the Topology Entity Id(s) of the OCUCP function instance(s) served by an NF Deployment.	type: URN multiplicity: 1..n isOrdered: False isUnique: True defaultValue: None	TE&IV
served-ocuupFunction	This represents the Topology Entity Id(s) of the OCUUP function instance(s) served by an NF Deployment.	type: URN multiplicity: 1..n isOrdered: False isUnique: True defaultValue: None	TE&IV
served-nearRTRICFunction	This represents the Topology Entity Id(s) of the Near-RT RIC function instance(s) served by an NF deployment.	type: URN multiplicity: 1..n isOrdered: False isUnique: True defaultValue: None	TE&IV

4.3.4 Namespace ORAN.SMO.TEIV

4.3.4.1 Overview

This section contains the commonly used O-RAN TE&IV basic types and abstractions for Topology Entities and Topology Relationships.

4.3.4.2 Imported associated information

4.3.4.2.1 Imported information entities and local labels

Imported information entities and local labels is not defined in the present version of the document.

4.3.4.2.2 Associated information entities and local labels

Associated information entities and local labels is not defined in the present version of the document.

4.3.4.3 Class diagram

4.3.4.3.1 Relationships

Not specified in the present version of the document.

4.3.4.3.2 Inheritance

Not specified in the present version of the document.

4.3.4.4 Class definitions

4.3.4.4.1 *TopologyEntity_*

4.3.4.4.1.1 Definitions

This abstract class represents the Topology Entities in the O-RAN TE&IV Information Models. All other <<TopologyEntityClass>> specified in this document must inherit from *TopologyEntity_* directly or indirectly.

4.3.4.4.1.2 Attributes

This class shall have the following attributes:

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
topologyEntityId	M	T	F	T	F
sourceIds	M	T	F	F	T

4.3.4.4.1.3 Attribute constraints

Not specified in the present version of the document.

4.3.4.4.1.4 Notifications

Not specified in the present version of the document.

4.3.4.4.1.5 State diagram

Not specified in the present version of the document.

4.3.4.4.2 *TopologyRel_*

4.3.4.4.2.1 Definitions

This abstract class represents the Topology Relationships between Topology Entities in the O-RAN TE&IV Information Models. All other <<TopologyRelationshipClass>> specified in this document must inherit from *TopologyRel_* directly or indirectly.

4.3.4.4.2.2 Attributes

This class shall have the following attributes:

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
topologyRelationshipId	M	T	F	T	F
sourceIds	M	T	F	F	T

4.3.4.4.2.3 Attribute constraints

Not specified in the present version of the document.

4.3.4.4.2.4 Notifications

Not specified in the present version of the document.

4.3.4.4.2.5 State diagram

Not specified in the present version of the document.

4.3.4.5 Attribute definitions

4.3.4.5.1 Attribute properties

The following table defines the properties of attributes specified in the present document.

Table 4.3.4.5.1-1: Attribute properties

Attribute Name	Documentation and Allowed Values	Properties	Source Domain
topologyEntityId	Unique identifier of the Topology Entity. This identity is represented using a URN notation format [14].	type: URN multiplicity: 1 isOrdered: N/A isUnique: True defaultValue: None	SMO/TE&IV
topologyRelationshipId	Unique identifier of the Topology Relationship. This identity is represented using a URN notation format [14].	type: URN multiplicity: 1 isOrdered: N/A isUnique: True defaultValue: None	SMO/TE&IV
sourceIds	Identities of the underlying topology resources represented by the Topology Entity or underlying topology resources participating in the Topology Relationship. The sourceIds are represented using a URN notation format [14]. A TE&IV consumer can use the identifiers to navigate to the source domain objects.	type: URN multiplicity: 1..N isOrdered: False isUnique: True defaultValue: None	e.g., O1

4.3.5 Namespace ORAN.SMO.TEIV.Physical

4.3.5.1 Namespace overview

This namespace contains the Topology Entities and Topology Relationship in the Physical domain, which comprises of the deployment aspects of O-RAN Physical Network Functions.

4.3.5.2 Imported associated information

4.3.5.2.1 Imported information entities and local labels

Table 4.3.5.2.1-1: Imported information entities and local labels

Label reference	Local label
TEC, *TopologyEntity_*	*TopologyEntity_*
TRC, *TopologyRel_*	*TopologyRel_*
3GPP TS 28.622[16], dataType, GeoCoordinate	GeoCoordinate

4.3.5.2.2 Associated information entities and local labels

Associated information entities and local labels is not defined in the present version of the document.

4.3.5.3 Class diagram

4.3.5.3.1 Relationships

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
skinparam linetype ortho
hide empty members
set namespaceSeparator none

namespace ORAN.SMO.TEIV.Physical {
class PhysicalAppliance_installedAt_Site
class PhysicalAppliance
class Site

PhysicalAppliance  "1..n\ninstalling-site\n\n" <---> "installed-physicalAppliance    \n1"
Site : > installedAt
PhysicalAppliance_installedAt_Site . (PhysicalAppliance, Site)
}
@enduml
```

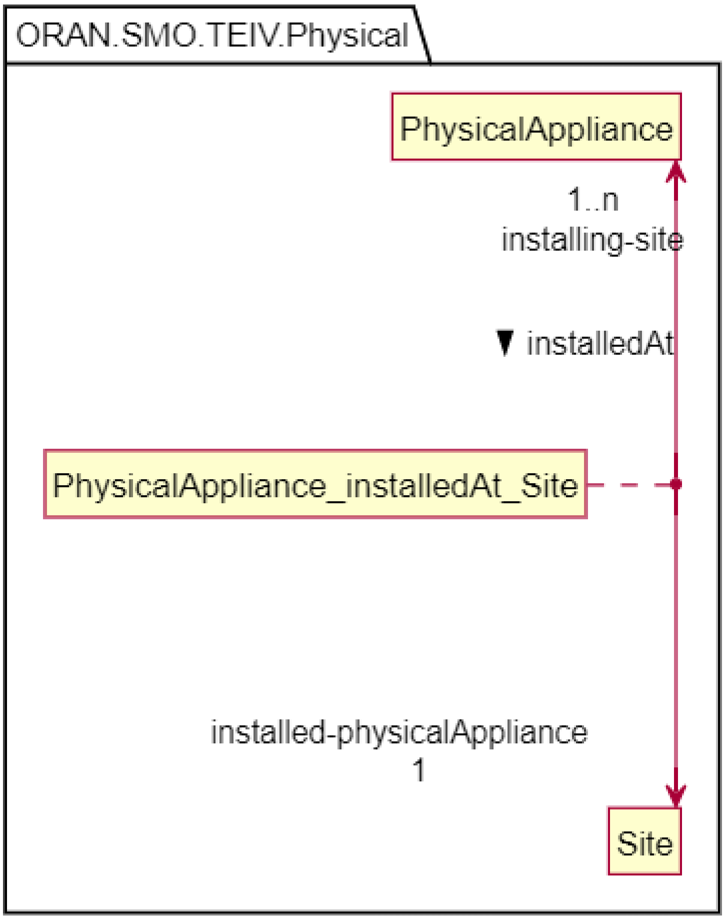

Figure 4.3.5.3.1-1: SMO TE&IV Physical Deployment relationship model

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members
set namespaceSeparator none
```

```
namespace ORAN.SMO.TEIV.Physical {

class PhysicalAppliance {
}

class Site {
}

class Site
PhysicalAppliance -> Site : PhysicalAppliance_installedAt_Site
}
@enduml
```

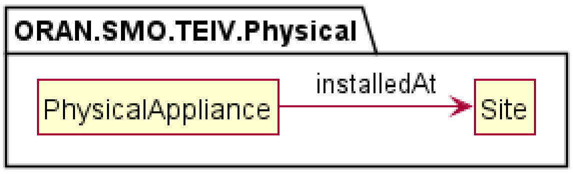

Figure 4.3.5.3.1-2: Simplified SMO TE&IV Physical Deployment relationship model

4.3.5.3.2 Inheritance

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members
set namespaceSeparator none

namespace ORAN.SMO.TEIV{
abstract class TopologyEntity_ {}
}

namespace ORAN.SMO.TEIV.Physical {
class Site
class PhysicalAppliance
}

TopologyEntity_ <|-- PhysicalAppliance
TopologyEntity_ <|-- Site

@enduml

@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members
set namespaceSeparator none

namespace ORAN.SMO.TEIV{
abstract class TopologyRel_ {}
}

namespace ORAN.SMO.TEIV.Physical {
class PhysicalAppliance_installedAt_Site
}

TopologyRel_ <|-- PhysicalAppliance_installedAt_Site

@enduml
```


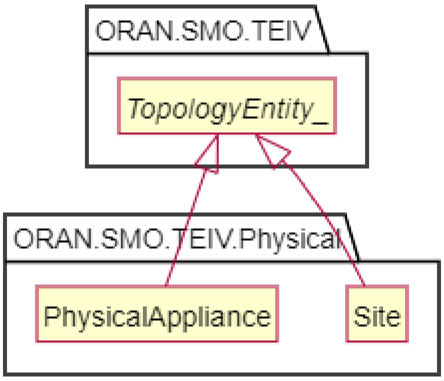

Figure 4.3.5.3.2-1: ORAN SMO TE&IV Physical Toplogy Entity inheritance view

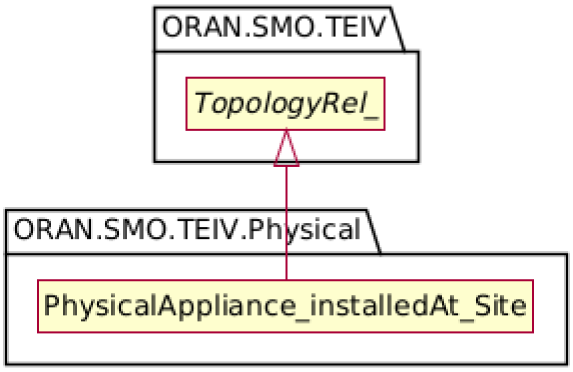

Figure 4.3.5.3.2-2: ORAN SMO TE&IV PhysicalAppliance_installedAt_Site inheritance view

4.3.5.4 Class definitions

4.3.5.4.1 PhysicalAppliance

4.3.5.4.1.1 Definitions

This class provides the TE&IV resource representation of PhysicalAppliance using the equivalent concept as defined in O-RAN.WG1.OAD [6] clause 5.

NOTE: The current version of this specification recommends that ORU be represented as a PhysicalAppliance with the applianceType as "ORU".

4.3.5.4.1.2 Attributes

The PhysicalAppliance TEC includes the topology attributes inherited from *TopologyEntity_* and have the following attributes:

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
vendorName	M	T	F	F	T
modelName	M	T	F	F	T
serialNumber	M	T	F	T	T
applianceType	M	T	F	F	T

4.3.5.4.1.3 Attribute constraints

Not specified in the present version of the document.

4.3.5.4.1.4 Notifications

Not specified in the present version of the document.

4.3.5.4.1.5 State diagram

Not specified in the present version of the document.

4.3.5.4.2 Site

4.3.5.4.2.1 Definitions

This class provides the TE&IV resource representation of physical Site using the similar concept as O-Cloud Site defined in O-RAN.WG6.O2-GA&P [7].

4.3.5.4.2.2 Attributes

The Site TEC includes the attributes inherited from *TopologyEntity_* and have the following attributes:

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
siteLocation	M	T	F	F	T
siteName	O	T	F	F	T

4.3.5.4.2.3 Attribute constraints

None.

4.3.5.4.2.4 Notifications

There is no notification defined.

4.3.5.4.2.5 State diagram

None.

4.3.5.4.3 GeoInformation <<dataType>>

4.3.5.4.3.1 Definitions

This data type provides the TE&IV resource representation of physical site location.

4.3.5.4.3.2 Attributes

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
address	O	T	F	F	T
coordinate	M	T	F	F	T

4.3.5.4.3.3 Attribute constraints

None.

4.3.5.4.4.4 Notifications

There is no notification defined.

4.3.5.4.4.5 State diagram

None.

4.3.5.4.4 PhysicalAppliance_installedAt_Site

4.3.5.4.4.1 Definitions

This class provides the Topology Relationship between the PhysicalAppliance and the Site Topology Entities as shown in figure 4.3.5.3.2-2. This class represents the relationship type of the Site installing the Physical Appliance.

4.3.5.4.4.2 Attributes

The Site TRC includes the attributes inherited from *TopologyRel_* and have the following attributes:

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
installing-site	M	T	F	F	T
installed-physicalAppliance	M	T	F	F	T

4.3.5.4.4.3 Attribute constraints

None.

4.3.5.4.4.4 Notifications

There is no notification defined.

4.3.5.4.4.5 State diagram

None.

4.3.5.5 Attribute definitions

4.3.5.5.1 Attribute properties

The following table defines the properties of attributes specified in the present document.

Table 4.3.5.5.1-1: Attribute properties

Attribute Name	Documentation and Allowed Values	Properties	Source Domain
vendorName	Name of the physical appliance vendor. allowedValues: Not applicable	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None	SMO
modelName	Name of the physical appliance model. allowedValues: Not applicable	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None	SMO
siteLocation	It indicates the location of the physical site. It describes Postal address and coordinate. allowedValues: Not applicable	type: GeoInformation multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None	SMO
siteName	Human readable name of the physical site as identified by the mobile network operator. allowedValues: Not applicable	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None	SMO

Attribute Name	Documentation and Allowed Values	Properties	Source Domain
address	Postal address of the location. allowedValues: Not applicable	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None	SMO
coordinate	The latitude, longitude and altitude of the GeoCoordinate, defined in clause 4.3.53 of 3GPP TS 28.622 [16]. allowedValues: Not applicable	type: GeoCoordinate multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None	SMO
installing-site	This represents the Topology Entity Id of the Site installing the PhysicalAppliance instance(s).	type: URN multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None	TE&IV
installed-physicalAppliance	This represents the Topology Entity Id(s) of the PhysicalAppliance instance(s) installed at the Site.	type: URN multiplicity: 1..n isOrdered: False isUnique: True defaultValue: None	TE&IV
serialNumber	Serial number of the appliance	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None	SMO
applianceType	Indicates the type of PhysicalAppliance.	type: String multiplicity: 0..1 isOrdered: N/A isUnique: N/A defaultValue:None	SMO

4.3.6 Namespace ORAN.SMO.TEIV.REL-Physical-RAN

4.3.6.1 Namespace overview

This namespace contains the relationship between specific Topology Entities of the ORAN.SMO.TEIV.RAN to ORAN.SMO.TEIV.Physical namespaces and which represents the physical deployment aspects of ORAN Network Functions.

4.3.6.2 Imported associated information

4.3.6.2.1 Imported information entities and local labels

Imported information entities and local labels is not defined in the present version of the document.

4.3.6.2.2 Associated information entities and local labels

Associated information entities and local labels is not defined in the present version of the document.

4.3.6.3 Class diagram

4.3.6.3.1 Relationships

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
```

```
hide empty members
set namespaceSeparator none

namespace ORAN.SMO.TEIV.RAN {
abstract class ORANNetworkFunction_

}
namespace ORAN.SMO.TEIV.REL-Physical-RAN {
abstract class PhysicalAppliance_REL_NF_

}
namespace ORAN.SMO.TEIV.Physical {
class PhysicalAppliance

}

PhysicalAppliance  "1..n" <-> "1..n" ORANNetworkFunction_
PhysicalAppliance_REL_NF_ "1" -[hidden]d- PhysicalAppliance
PhysicalAppliance_REL_NF_ -[hidden]d- ORANNetworkFunction_
PhysicalAppliance_REL_NF_ . (PhysicalAppliance, ORANNetworkFunction_)
@enduml
```

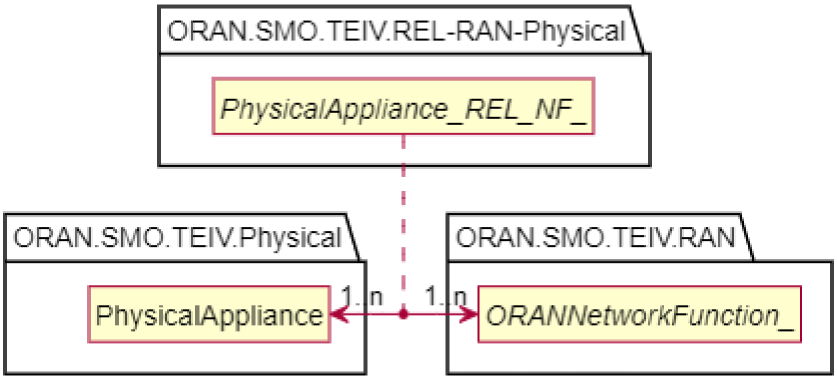

Figure 4.3.6.3.1-1: SMO TE&IV Physical-RAN relationship model

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
skinparam nodesep 120
hide empty members
set namespaceSeparator none

namespace ORAN.SMO.TEIV.Physical {
class PhysicalAppliance
}

namespace ORAN.SMO.TEIV.RAN {
abstract ORANNetworkFunction_
}



PhysicalAppliance <-right->  ORANNetworkFunction_: \t<b><i>PhysicalAppliance_REL_NF_


@enduml
```

Figure 4.3.6.3.1-2: Simplified SMO TE&IV Physical-RAN relationship model

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members
set namespaceSeparator none

namespace ORAN.SMO.TEIV.RAN {
class ODUFunction{
    }

}
namespace ORAN.SMO.TEIV.REL-Physical-RAN {
class PhysicalAppliance_serves_ODUFunction{
    }
}
namespace ORAN.SMO.TEIV.Physical {
class PhysicalAppliance{
    }
}

PhysicalAppliance  " \n1..n\nserved-oduFunction" <---> "serving-
physicalAppliance\n1..n\n\n   " ODUFunction: > serves
PhysicalAppliance_serves_ODUFunction . (PhysicalAppliance, ODUFunction)
@enduml
```


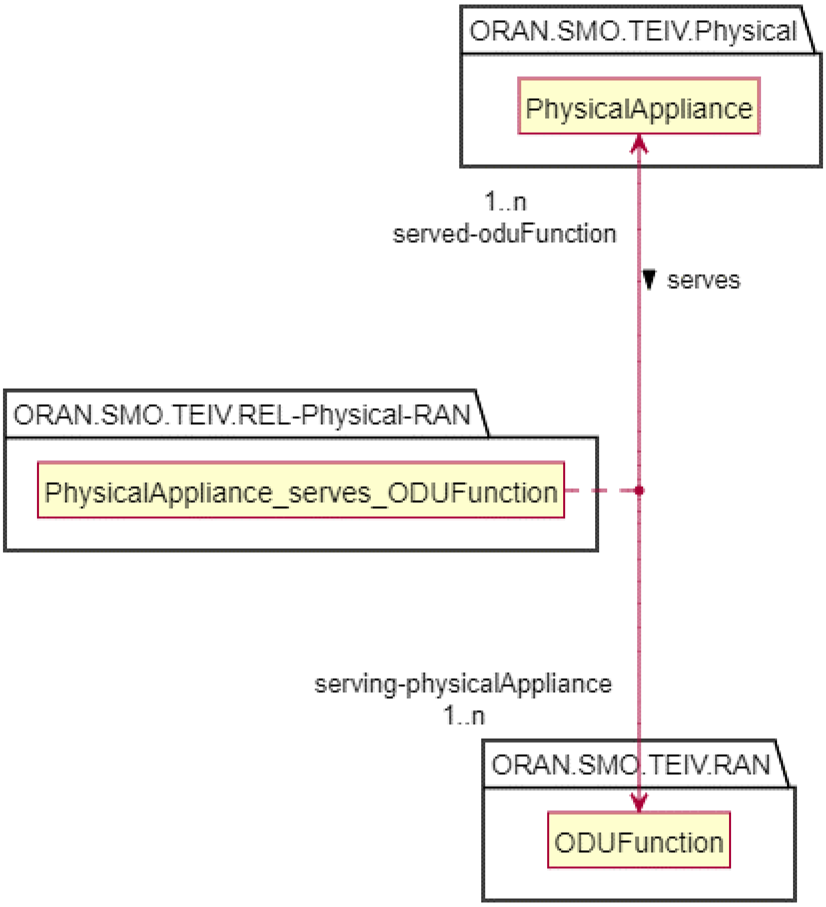

Figure 4.3.6.3.1-3: PhysicalAppliance and ODUFunction relationship model

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members
set namespaceSeparator none

namespace ORAN.SMO.TEIV.RAN {
class OCUUPFunction{
    }

}
namespace ORAN.SMO.TEIV.REL-Physical-RAN {
class PhysicalAppliance_serves_OCUUPFunction{
    }
}
namespace ORAN.SMO.TEIV.Physical {
class PhysicalAppliance{
    }
}

PhysicalAppliance  " \n1..n\nserved-ocuupFunction" <---> "serving-
physicalAppliance\n1..n\n\n   " OCUUPFunction: > serves
PhysicalAppliance_serves_OCUUPFunction . (PhysicalAppliance, OCUUPFunction)
@enduml
```


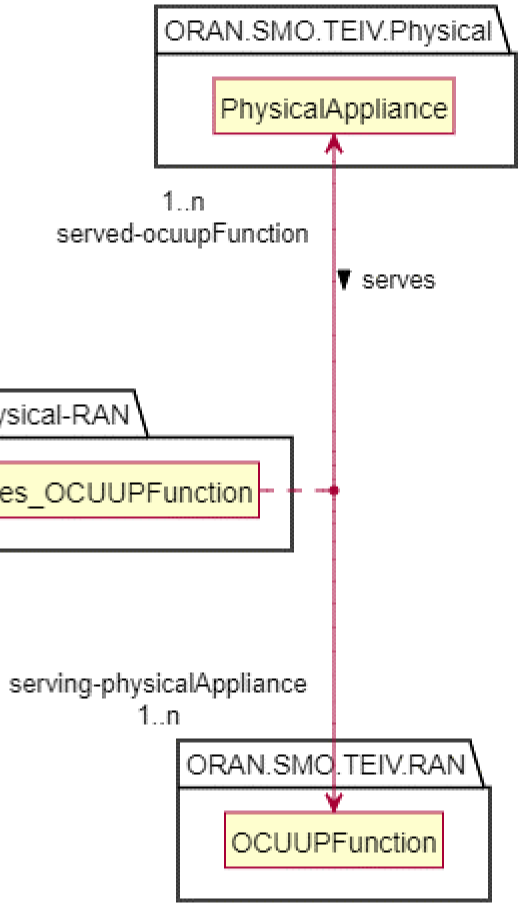

Figure 4.3.6.3.1-4: PhysicalAppliance and OCUUPFunction relationship model

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members
set namespaceSeparator none

namespace ORAN.SMO.TEIV.RAN {
class OCUCPFunction{
    }

}
namespace ORAN.SMO.TEIV.REL-Physical-RAN {
class PhysicalAppliance_serves_OCUCPFunction{
    }
}
namespace ORAN.SMO.TEIV.Physical {
class PhysicalAppliance{
    }
}

PhysicalAppliance  " \n1..n\nserved-ocucpFunction" <---> "serving-
physicalAppliance\n1..n\n\n   " OCUCPFunction: > serves
PhysicalAppliance_serves_OCUCPFunction . (PhysicalAppliance, OCUCPFunction)
@enduml
```


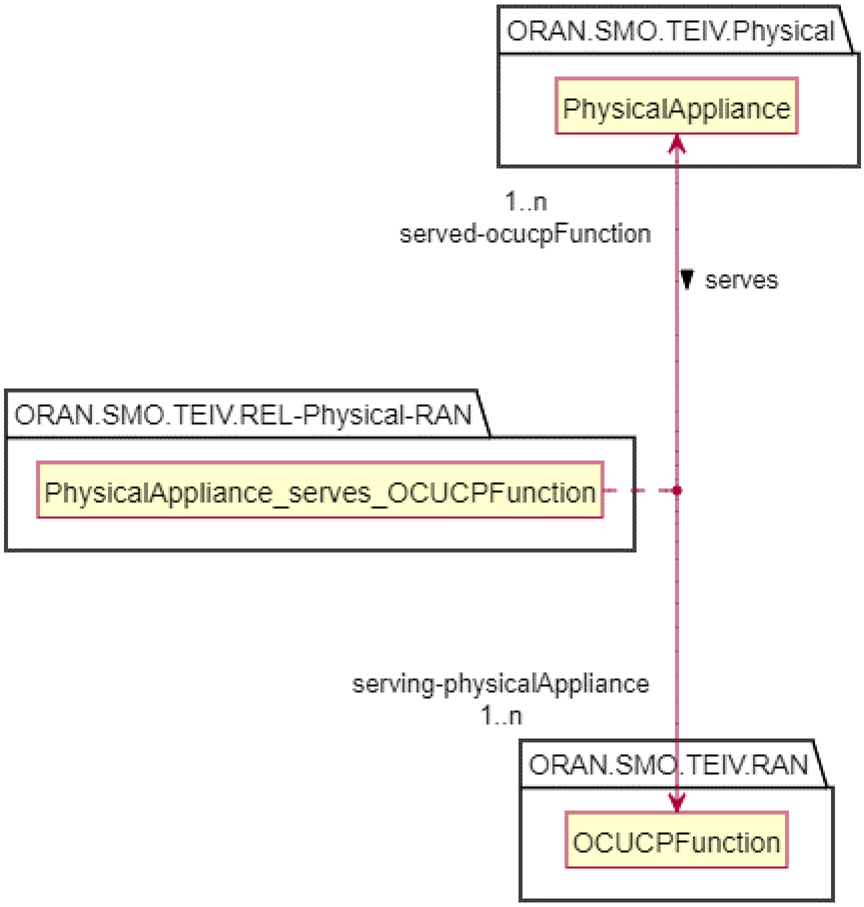

Figure 4.3.6.3.1-5: PhysicalAppliance and OCUCPFunction relationship model

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members
set namespaceSeparator none

namespace ORAN.SMO.TEIV.RAN {
class NearRTRICFunction{
    }

}
namespace ORAN.SMO.TEIV.REL-Physical-RAN {
class PhysicalAppliance_serves_NearRTRICFunction{
    }
}
namespace ORAN.SMO.TEIV.Physical {
class PhysicalAppliance{
    }
}

PhysicalAppliance  " \n1..n\nserved-nearRTRICFunction" <---> "serving-
physicalAppliance\n1..n\n\n   " NearRTRICFunction: > serves
PhysicalAppliance_serves_NearRTRICFunction . (PhysicalAppliance, NearRTRICFunction)
@enduml
```


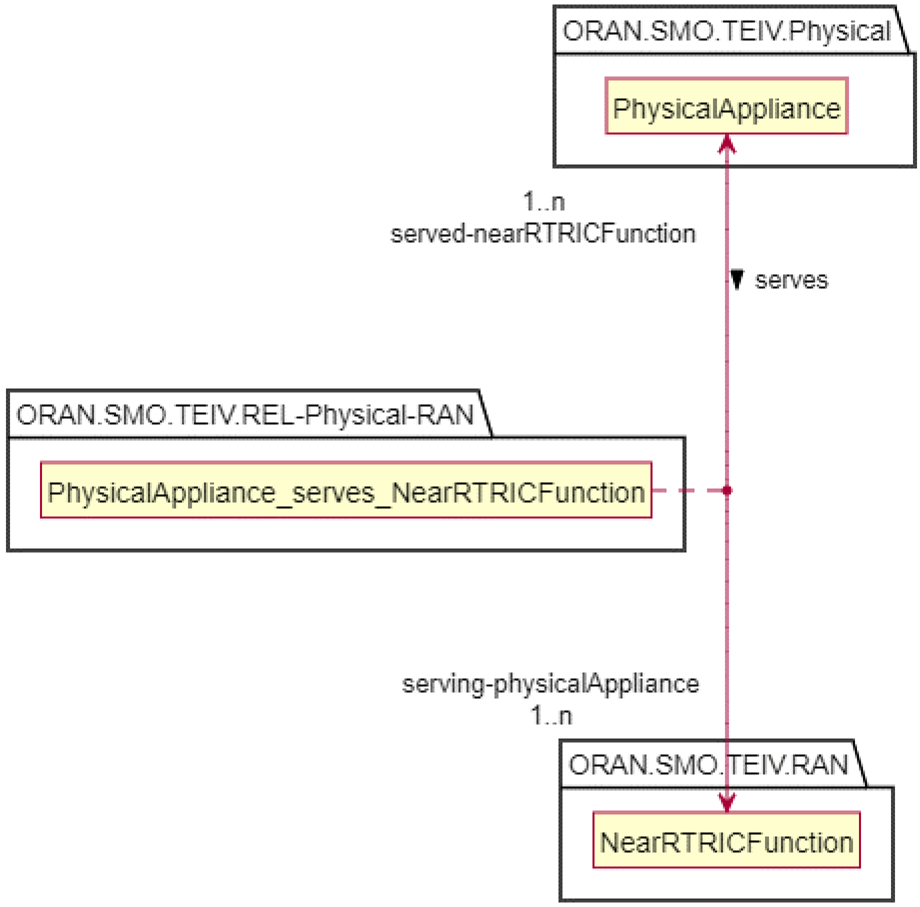

Figure 4.3.6.3.1-6: PhysicalAppliance and NearRTRICFunction relationship model

4.3.6.3.2 Inheritance

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members
set namespaceSeparator none

namespace ORAN.SMO.TEIV{
abstract class TopologyRel_ {}
}

namespace ORAN.SMO.TEIV.REL-Physical-RAN {
abstract class PhysicalAppliance_REL_NF_ {}

TopologyRel_ <|-- PhysicalAppliance_REL_NF_
PhysicalAppliance_REL_NF_ <|-- PhysicalAppliance_serves_ODUFunction
PhysicalAppliance_REL_NF_ <|-- PhysicalAppliance_serves_OCUCPFunction
PhysicalAppliance_REL_NF_ <|-- PhysicalAppliance_serves_OCUUPFunction
PhysicalAppliance_REL_NF_ <|-- PhysicalAppliance_serves_NeartRTRICFunction
}
@enduml
```


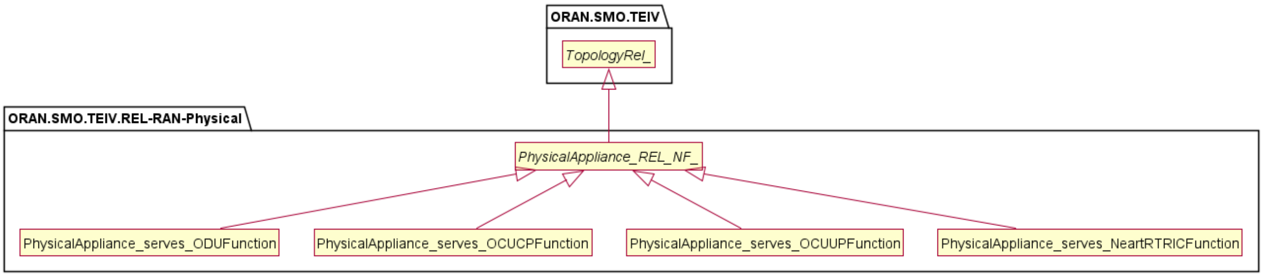

Figure 4.3.6.3.2-1: ORAN SMO TE&IV RAN Physical inheritance view

4.3.6.4 Class definitions

4.3.6.4.1 PhysicalAppliance_serves_ODUFunction

4.3.6.4.1.1 Definitions

This class provides the Topology Relationship between the PhysicalAppliance and the ODUFunction Topology Entities as shown in figure 4.3.6.3.1-3. This class represents the relationship type of the PhysicalAppliance serving the functionality of the ODU function.

4.3.6.4.1.2 Attributes

The PhysicalAppliance_serves_ODUFunction TRC includes the attributes inherited from *TopologyRel_* and have the following attributes:

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
served-oduFunction	M	T	F	F	T
serving-physicalAppliance	M	T	F	F	T

4.3.6.4.1.3 Attribute constraints

None.

4.3.6.4.1.4 Notifications

None.

4.3.6.4.1.5 State diagram

None.

4.3.6.4.2 PhysicalAppliance_serves_OCUUPFunction

4.3.6.4.2.1 Definitions

This class provides the Topology Relationship between the PhysicalAppliance and the OCUUPFunction Topology Entities as shown in figure 4.3.6.3.1-4. This class represents the relationship type of the PhysicalAppliance serving the functionality of the OCUUP function.

4.3.6.4.2.2 Attributes

The PhysicalAppliance_serves_OCUUPFunction TRC includes the attributes inherited from *TopologyRel_* and have the following attributes:

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
served-ocuupFunction	M	T	F	F	T
serving-physicalAppliance	M	T	F	F	T

4.3.6.4.2.3 Attribute constraints

None.

4.3.6.4.2.4 Notifications

None.

4.3.6.4.2.5 State diagram

None.

4.3.6.4.3 PhysicalAppliance_serves_OCUCPFunction

4.3.6.4.3.1 Definitions

This class provides the Topology Relationship between the PhysicalAppliance and the OCUCPFunction Topology Entities as shown in figure 4.3.6.3.1-5. This class represents the relationship type of the PhysicalAppliance serving the functionality of the OCUCP function.

4.3.6.4.3.2 Attributes

The PhysicalAppliance_serves_OCUCPFunction TRC includes the attributes inherited from *TopologyRel_* and have the following attributes:

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
served-ocucpFunction	M	T	F	F	T
serving-physicalAppliance	M	T	F	F	T

4.3.6.4.3.3 Attribute constraints

None.

4.3.6.4.3.4 Notifications

None.

4.3.6.4.3.5 State diagram

None.

4.3.6.4.4 PhysicalAppliance_serves_NearRTRICFunction

4.3.6.4.4.1 Definitions

This class provides the Topology Relationship between the PhysicalAppliance and the NearRTRICFunction Topology Entities as shown in figure 4.3.6.3.1-6. This class represents the relationship type of the PhysicalAppliance serving the functionality of the Near-RTRIC function.

4.3.6.4.4.2 Attributes

The PhysicalAppliance_serves_NearRTRICFunction TRC includes the attributes inherited from *TopologyRel_* and have the following attributes:

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
served-nearRTRICFunction	M	T	F	F	T
serving-physicalAppliance	M	T	F	F	T

4.3.6.4.4.3 Attribute constraints

None

4.3.6.4.4.4 Notifications

None.

4.3.6.4.4.5 State diagram

None.

4.3.6.4.5 *PhysicalAppliance_REL_NF_*

4.3.6.4.5.1 Definitions

This abstract class is provided for sub-classing only and is used to generalize the relationship classes in this namespace.

4.3.6.4.5.2 Attributes

None.

4.3.6.4.5.3 Attribute constraints

None.

4.3.6.4.5.4 Notifications

None.

4.3.6.4.5.5 State diagram

None.

4.3.6.4 Attribute definitions

4.3.6.4.1 Attribute properties

The following table defines the properties of attributes specified in the present document.

Table 4.3.6.4.1-1: Attribute properties

Attribute Name	Documentation and Allowed Values	Properties	Source Domain
serving-physicalAppliance	This represents the Topology Entity Id(s) of the NF PhysicalAppliance instance(s) serving O-RAN Network Function(s).	type: URN multiplicity: 1..n isOrdered: False isUnique: True	TE&IV

Attribute Name	Documentation and Allowed Values	Properties	Source Domain
		defaultValue: None	
served-oduFunction	This represents the Topology Entity Id(s) of the ODU function instance(s) served by the PhysicalAppliance.	type: URN multiplicity: 1..n isOrdered: False isUnique: True defaultValue: None	TE&IV
served-ocuupFunction	This represents the Topology Entity Id(s) of the OCUUP function instance(s) served by the PhysicalAppliance.	type: URN multiplicity: 1..n isOrdered: False isUnique: True defaultValue: None	TE&IV
served-ocucpFunction	This represents the Topology Entity Id(s) of the OCUCP function instance(s) served by the PhysicalAppliance.	type: URN multiplicity: 1..n isOrdered: False isUnique: True defaultValue: None	TE&IV
served-nearRTRICFunction	This represents the Topology Entity Id(s) of the Near-RTRIC function instance(s) served by the PhysicalAppliance.	type: URN multiplicity: 1..n isOrdered: False isUnique: True defaultValue: None	TE&IV

4.4 TE&IV Service Operations

4.4.1 Introduction

This clause provides the stage 2 definitions of TE&IV Service Operations for exposing Topology Entities, Topology Relationships and their attributes.

The information model in Figure 4.4.1-1 depicts the model view of the entities used in the TE&IV Service Operations initiated by TE&IV Service Consumers to retrieve the Domains in Topology, Topology Entities and Topology Relationships.

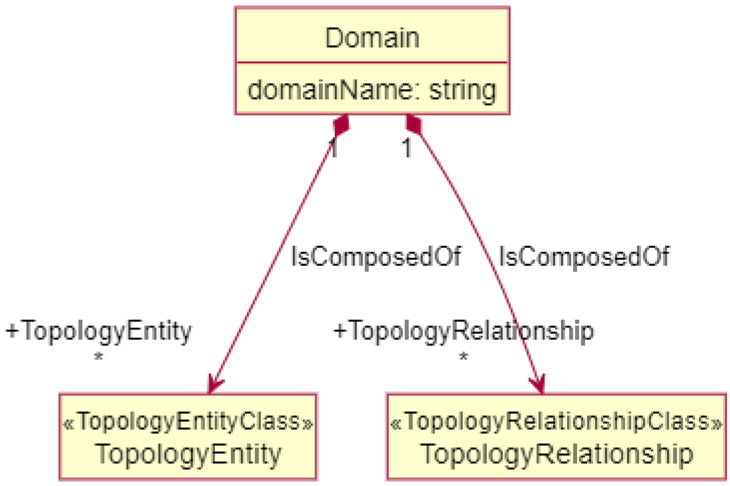

Figure 4.4.1-1: ORAN SMO TE&IV Topology Domain, Topology Entity and Topology Relationship Model View

The Figure 4.4.1-2 shows the allowed inter-domain relationships. It depicts subclasses of the Domain concept comprising two distinct categories: Relationship and Entity. An Entity Domain may contain both Entities and Relationships, however both aSide and bSide of such relationships shall exist in that domain. A Relationship Domain encapsulates Relationships to Entities in one or more Entity Domains. The Relationship Domain should not have any Entities.

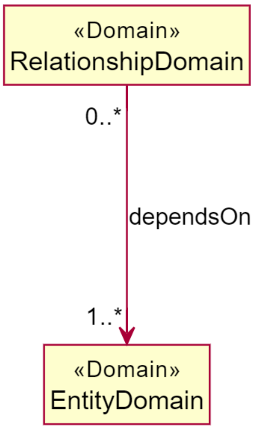

Figure 4.4.1-2: ORAN SMO TE&IV Domain Dependency Model View

4.4.2 Operations and Notification

4.4.2.1 getAllDomains operation

4.4.2.1.1 Description

This operation is invoked by the TE&IV Service Consumer to retrieve all the available Topology Domains.

4.4.2.1.2 Input parameters

Parameter Name	S	Description
Accept	M	This parameter specifies the response media types that are acceptable.
offsetParam	O	This query parameter allows you to omit a specified number of entries before the beginning of the result set for pagination.
limitParam	O	The query parameter provides to limit the number of entries returned for a request for pagination.

4.4.2.1.3 Output parameters

Parameter name	S	Description
Domains	M	A list of Topology Domains and the links to its EntityTypes and RelationshipTypes.

4.4.2.1.4 Results

In case of success, all the available Topology Domains are returned. In case of failure, an appropriate error response may be provided.

4.4.2.2 getTopologyEntityTypes operation

4.4.2.2.1 Description

This operation is invoked by the TE&IV Service Consumer to get all the available Topology Entity types within a domain.

4.4.2.2.2 Input parameters

Parameter Name	S	Description
Accept	M	This parameter specifies the response media types that are acceptable.
domainName	M	This path parameter specifies the name of the Topology Domain.
offsetParam	O	This query parameter allows you to omit a specified number of entries before the beginning of the result set for pagination.
limitParam	O	The query parameter provides to limit the number of entries returned for a request for pagination.

4.4.2.2.3 Output parameters

Parameter name	S	Description
EntityTypes	M	A list of links to the Topology Entities within the Topology Domain.

4.4.2.2.4 Results

In case of success, all the available Topology Entity types in domain name are returned. In case of failure, an appropriate error response may be provided.

4.4.2.3 getTopologyByEntityTypeName operation

4.4.2.3.1 Description

This operation is invoked by the TE&IV Service Consumer to get all Topology Entity instances of a specific Topology Entity type.

4.4.2.3.2 Input parameters

Parameter Name	S	Description
Accept	M	This parameter specifies the response media types that are acceptable.
domainName	M	This path parameter specifies the name of the Topology Domain.
entityTypeName	M	This path parameter specifies the name of a Topology Entity in a Topology Domain.
targetFilter	O	This query parameter specifies the entity type and attributes to be returned in the REST response.
scopeFilter	O	This query parameter specifies the attributes to match on for specific Topology Entities for which the data is to be produced.
offsetParam	O	This query parameter allows you to omit a specified number of entries before the beginning of the result set for pagination.
limitParam	O	The query parameter provides to limit the number of entries returned for a request for pagination.

4.4.2.3.3 Output parameters

Parameter name	S	Description
Entities	M	A list of links to the data model for schema definition of Topology Entities.

4.4.2.3.4 Results

In case of success, all the available instances of a specific topology entity type are returned. In case of failure, an appropriate error response may be provided.

4.4.2.4 getTopologyById operation

4.4.2.4.1 Description

This operation is invoked by the TE&IV Service Consumer to get a specific Topology Entity instance of a Topology Entity type.

4.4.2.4.2 Input parameters

Parameter Name	S	Description
Accept	M	This parameter specifies the response media types that are acceptable.
domainName	M	This path parameter specifies the name of the Topology Domain.
entityTypeName	M	This path parameter specifies the name of a Topology Entity in a Topology Domain.
entityId	M	The path parameter specifies the Identifier of a Topology Entity instance.

4.4.2.4.3 Output parameters

Parameter name	S	Description
EntityInstance	M	Encapsulated object reference to the data model for schema definition of Topology Entities.

4.4.2.4.4 Results

In case of success, an object referencing to the data model schema definition of Topology Entity instance of a specific Topology Entity type name is returned. In case of failure, an appropriate error response may be provided.

4.4.2.5 getAllRelationshipsForEntityId operation

4.4.2.5.1 Description

This operation is invoked by the TE&IV Service Consumer to get all relationships for a specific Topology Entity instance of a Topology Entity type.

4.4.2.5.2 Input parameters

Parameter Name	S	Description
Accept	M	This parameter specifies the response media types that are acceptable.
domainName	M	This path parameter specifies the name of the Topology Domain.
entityTypeName	M	This path parameter specifies the name of a Topology Entity in a Topology Domain.
entityId	M	The path parameter specifies the Identifier of a Topology Entity instance.
targetFilter	O	This query parameter specifies the entity type and relationship to be returned in the REST response.
scopeFilter	O	This query parameter specifies the attributes to match on for specific Topology Entity relationship for which the data is to be produced.
offsetParam	O	This query parameter allows you to omit a specified number of entries before the beginning of the result set for pagination.
limitParam	O	The query parameter provides to limit the number of entries returned for a request for pagination.

4.4.2.5.3 Output parameters

Parameter name	S	Description
Relationships	M	Encapsulated object reference to the data model for schema definition of Topology Relationships.

4.4.2.5.4 Results

In case of success, an object referencing to the data model schema definition of Topology Relationships of a specific Topology Entity type name is returned. In case of failure, an appropriate error response may be provided.

4.4.2.6 getTopologyRelationshipTypes operation

4.4.2.6.1 Description

This operation is invoked by the TE&IV Service Consumer to retrieve all the available Topology Relationship types.

4.4.2.6.2 Input parameters

Parameter Name	S	Description
Accept	M	This parameter specifies the response media types that are acceptable.
domainName	M	This path parameter specifies the name of the Topology Domain.
offsetParam	O	This query parameter allows you to omit a specified number of entries before the beginning of the result set for pagination.
limitParam	O	The query parameter provides to limit the number of entries returned for a request for pagination.

4.4.2.6.3 Output parameters

Parameter name	S	Description
RelationshipTypes	M	A list of links to the Topology Relationships.

4.4.2.6.4 Results

In case of success, all the available Topology Relationship types are returned. In case of failure, an appropriate error response may be provided.

4.4.2.7 getRelationshipsByType operation

4.4.2.7.1 Description

This operation is invoked by the TE&IV Service Consumer to get all the available Topology Relationships of a specific relationship type name.

4.4.2.7.2 Input parameters

Parameter Name	S	Description
Accept	M	This parameter specifies the response media types that are acceptable.
domainName	M	This path parameter specifies the name of the Topology Domain.
relationshipTypeName	M	This path parameter specifies the name of a Topology Relationship in a Topology Domain.
targetFilter	O	This query parameter specifies the entity type and attributes to be returned in the REST response.
scopeFilter	O	This query parameter specifies the attributes to match on for specific Topology Entities for which the data is to be produced.
offsetParam	O	This query parameter allows you to omit a specified number of entries before the beginning of the result set for pagination.
limitParam	O	The query parameter provides to limit the number of entries returned for a request for pagination.

4.4.2.7.3 Output parameters

Parameter name	S	Description
Relationships	M	Encapsulated object reference to the data model for schema definition of Topology Relationships.

4.4.2.7.4 Results

In case of success, all the available Topology Relationships of a specific relationship type name are returned. In case of failure, an appropriate error response may be provided.

4.4.2.8 getRelationshipById operation

4.4.2.8.1 Description

This operation is invoked by the TE&IV Service Consumer to get a specific Topology Relationship instance of a Topology Relationship type.

4.4.2.8.2 Input parameters

Parameter Name	S	Description
Accept	M	This parameter specifies the response media types that are acceptable.
domainName	M	This path parameter specifies the name of the Topology Domain.
relationshipTypeName	M	This path parameter specifies the name of a Topology Relationship in a Topology Domain.
relationshipId	M	This path parameter specifies the identifier of a Topology Relationship instance.

4.4.2.8.3 Output parameters

Parameter name	S	Description
Relationship	M	Encapsulated object reference to the data model for schema definition of Topology Relationship.

4.4.2.8.4 Results

In case of success, a specific Topology Relationship instance of a Topology Relationship type is returned. In case of failure, an appropriate error response may be provided.

4.4.2.9 getSchemas operation

4.4.2.9.1 Description

This operation is invoked by the TE&IV Service Consumer to retrieve all the available Topology Model Schemas.

4.4.2.9.2 Input parameters

Parameter Name	S	Description
Accept	M	This parameter specifies the response media types that are acceptable.
offsetParam	O	This query parameter allows you to omit a specified number of entries before the beginning of the result set for pagination.
limitParam	O	The query parameter provides to limit the number of entries returned for a request for pagination.
domain	O	This query parameter allows you to specify the desired domain

4.4.2.9.3 Output parameters

Parameter name	S	Description
Schemas	M	A list of Topology Model Schemas

4.4.2.9.4 Results

In case of success, all the available Topology Model Schemas are returned. In case of failure, an appropriate error response may be provided.

4.4.2.10 getSchemaByName operation

4.4.2.10.1 Description

This operation is invoked by the TE&IV Service Consumer to get the schema by name.

4.4.2.10.2 Input parameters

Parameter Name	S	Description
Accept	M	This parameter specifies the response media types that are acceptable.
schemaName	M	This path parameter specifies the name of the Topology Model Schema.
offsetParam	O	This query parameter allows you to omit a specified number of entries before the beginning of the result set for pagination.
limitParam	O	The query parameter provides to limit the number of entries returned for a request for pagination.

4.4.2.10.3 Output parameters

Parameter name	S	Description
string	M	Content of the Topology Model Schema

4.4.2.10.4 Results

In case of success, schema content is returned. In case of failure, an appropriate error response may be provided.

4.5 TE&IV User Defined Data

4.5.1 Introduction

TE&IV user defined data is used for enriching Topology Entities and Topology Relationships. Such data can be added to existing Topology Entities and Relationships which enables flexibility for TE&IV Service Consumers in its usage of the service operations defined in clause 4.4. The user defined data are declared in separate schemas and are added through a REST API which shall be defined in the TE&IV API [17]. The following clauses provide the definitions for the types of TE&IV user defined data.

4.5.2 Classifiers

Classifiers permit the association of a well defined user specified string with a Topology Entity and/or Topology Relationship. Classifiers are declared in a schema added through a REST API [17]. When the schema is successfully created and validated, the user can assign the classifiers to the selected entities and/or relationships.

4.5.2.1 Definitions

This class provides the representation of Classifiers.

4.5.2.2 Attributes

Classifiers have the following attributes:

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
operation	M	T	F	F	T
classifierNames	M	T	F	F	T
topologyEntityIds	M	T	F	F	T
topologyRelationshipIds	M	T	F	F	T

4.5.2.3 Attribute constraints

None.

4.5.2.4 Notifications

There is no notification defined.

4.5.2.5 State diagram

None.

4.5.3 Decorators

Decorators are key-value pairs that can be associated with Topology Entities and/or Topology Relationships. The type of the value is defined by the user in the schema which declares the decorator. Decorators are declared in a schema added through a REST API [17]. When the schema is successfully created and validated, the user can assign the decorator to the selected entities and/or relationships.

4.5.3.1 Definitions

This class provides the representation of Decorators.

4.5.3.2 Attributes

Decorators have the following attributes:

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
operation	M	T	F	F	T
decoratorTags	M	T	F	F	T
topologyEntityIds	M	T	F	F	T
topologyRelationshipIds	M	T	F	F	T

4.5.2.3 Attribute constraints

None.

4.5.2.4 Notifications

There is no notification defined.

4.5.2.5 State diagram

None.

4.5.4 Attribute properties

Table 4.5.4.-1: Attribute properties

Attribute Name	Documentation and Allowed Values	Properties	Source Domain
operation	Specifies whether a classifier or decorator is being added or removed. Allowed values: merge, delete Merge: Defined classifiers or decorators can be assigned to specified Topology Entities and Topology Relationships in a single request. Delete: Defined classifiers or decorators can be removed from specified Topology Entities and Topology Relationships in a single request.	type: ENUM multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None	
classifierNames	List of tags to be assigned to Topology Entities and/or Topology Relationships	type: String multiplicity: 1..N isOrdered: N/A isUnique: N/A defaultValue: None	
decoratorTags	Key-value pairs to be assigned to Topology Entities and/or Topology Relationships. The content of this attribute is a list of attributeName-attributeValue pairs. Allowed values: Value must be a simple data type (string, integer, or Boolean)	type: AttributeValuePair multiplicity: 1..N isOrdered: N/A isUnique: N/A defaultValue: None	
topologyEntityIds	List of topologyEntityId. topologyEntityId is defined in clause 4.3.4.5.1.	type: URN multiplicity: 1..N isOrdered: N/A isUnique: N/A defaultValue: None	

4.5.5 Service Operations for User Defined Data

This clause provides the stage 2 definitions of TE&IV Service Operations for exposing Topology Entities, Topology Relationships using User Defined Data.

4.5.5.1 updateClassifier operation

4.5.5.1.1 Description

This operation is invoked by the TE&IV Service Consumer to merge or delete the classifiers for the available Topology Entities and Relationships.

4.5.5.1.2 Input parameters

Parameter Name	S	Description
Accept	M	This parameter specifies the response media types that are acceptable.
Content-Type	M	This parameter specifies the media types of the resource.

4.5.5.1.3 Output parameters

None

4.5.4.1.4 Results

In case of success, the classifier is assigned to or removed from all indicated entities and relationships. In case of failure, an appropriate error response may be provided.

4.5.5.2 updateDecorator operation

4.5.5.2.1 Description

This operation is invoked by the TE&IV Service Consumer to merge or delete the decorators for the available Topology Entities and Relationships.

4.5.5.2.2 Input parameters

Parameter Name	S	Description
Accept	M	This parameter specifies the response media types that are acceptable.
Content-Type	M	This parameter specifies the media types of the resource.

4.5.5.2.3 Output parameters

None.

4.5.5.2.4 Results

In case of success, all indicated entities and relationships are updated with the decorator. In case of failure, an appropriate error response may be provided.

4.5.5.3 getUserDefinedSchema operation

4.5.5.3.1 Description

This operation is invoked by the TE&IV Service Consumer to discover the available user defined schema.

4.5.5.3.2 Input parameters

Parameter Name	S	Description
Accept	M	This parameter specifies the response media types that are acceptable.
offsetParam	O	This query parameter allows you to omit a specified number of entries before the beginning of the result set for pagination.
limitParam	O	The query parameter provides to limit the number of entries returned for a request for pagination.

4.5.5.3.3 Output parameters

Parameter Name	S	Description
UserDefinedSchemas	M	A list of user defined schemas.

4.5.5.3.4 Results

In case of success, all the available user defined schemas are returned. In case of failure, an appropriate error response may be provided.

4.5.5.4 createUserDefinedSchema operation

4.5.5.4.1 Description

This operation is invoked by the TE&IV Service Consumer to create new user defined schemas.

4.5.5.4.2 Input parameters

Parameter Name	S	Description
Accept	M	This parameter specifies the response media types that are acceptable.
Content-Type	M	This parameter specifies the media types of the resource.
MultipartFile	M	Multipart file containing the user defined schema to be created.

4.5.5.4.3 Output parameters

Parameter Name	S	Description
UserDefinedSchema	M	User defined schema with link to its content.

4.5.5.4.4 Results

In case of success, the new user defined schema is created. In case of failure, an appropriate error response may be provided.

4.5.5.5 deleteUserDefinedSchema operation

4.5.5.5.1 Description

This operation is invoked by the TE&IV Service Consumer to delete a user defined schema.

4.5.5.5.2 Input parameters

Parameter Name	S	Description
Accept	M	This parameter specifies the response media types that are acceptable.
schemaName	M	This path parameter specifies the name of the user defined schema.

4.5.5.5.3 Output parameters

None.

4.5.5.5.4 Results

In case of success, the specified user defined schema is deleted. In case of failure, an appropriate error response may be provided.

4.5.5.6 getUserDefinedSchemaByName operation

4.5.5.6.1 Description

This operation is invoked by the TE&IV Service Consumer to get a specific user defined schema by name.

4.5.5.6.2 Input parameters

Parameter Name	S	Description
Accept	M	This parameter specifies the response media types that are acceptable.
Content-Type	M	This parameter specifies the media types of the resource.
schemaName	M	This path parameter specifies the name of the user defined schema.

4.5.5.6.3 Output parameters

Parameter Name	S	Description
string	M	User defined schema content.

4.5.5.6.4 Results

In case of success, the content of the user defined schema is returned. In case of failure, an appropriate error response may be provided.

Annex A (Informative): Model views

A.1 Model View: O-RAN TE&IV Network Function Deployment View (informative)

The TE&IV Network Function Deployment model view provides the basic constructs of the Topology Entities and Topology Relationship to represent the topology information related to the physical and cloud deployment aspects of the O-RAN Network Functions. The physical network function deployment view is shown in Figure A.1-1 and the cloud network function deployment view is shown in Figure A.1-2.

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members
set namespaceSeparator none
package "TE&IV Physical Network Function Deployment View" {

class PhysicalAppliance

abstract class ORANNetworkFunction_
class Site

note "has location info." as N1
note "Represents a Physical hardware which is used to realise Network Functions" as P1
}

PhysicalAppliance .r. P1

Site .r. N1

ORANNetworkFunction_ "*" <-- "0..n" PhysicalAppliance : serves
PhysicalAppliance "1..*" --> "0..1" Site : installedAt

@enduml
```


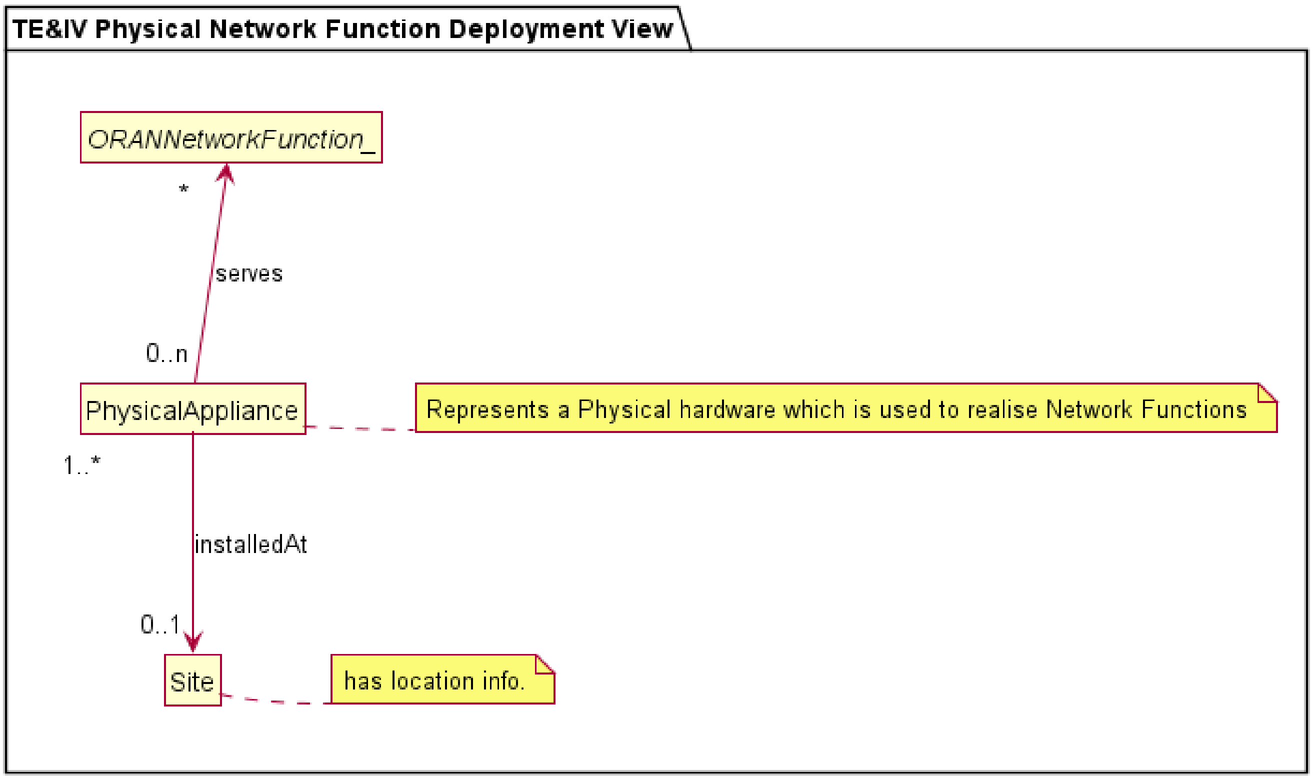

Figure A.1-1: ORAN TE&IV Physical NF Deployment View

```
@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members
set namespaceSeparator none
package "TE&IV Cloud Network Function Deployment View" {

class CloudifiedNF
class NFDeployment

abstract class ORANNetworkFunction_

class OCloudNamespace
class NodeCluster
class OCloudSite

note "has location info." as N1
}

ORANNetworkFunction_ "1..n" <-- "0..n" NFDeployment : serves

CloudifiedNF"1" --> "1..*" NFDeployment: comprizes
NFDeployment "1..* " --> "1..*" OCloudNamespace : deployedOn
OCloudNamespace "1..* " --> "1" NodeCluster : deployedOn
NodeCluster "1..* " --> "1..*" OCloudSite : locatedAt

OCloudSite .r. N1
@enduml
```


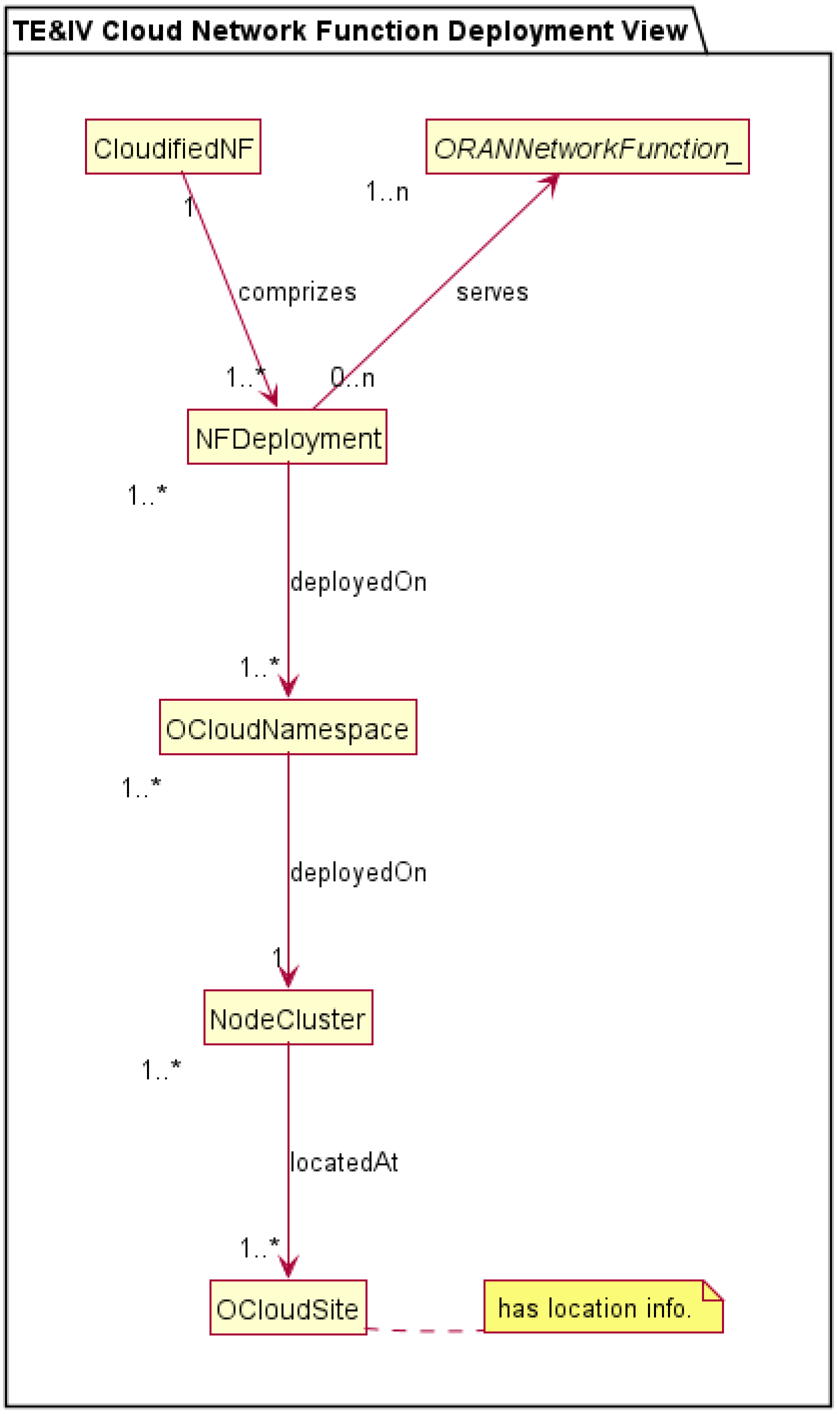

Figure A.1-2: ORAN TE&IV Cloud NF Deployment View

Annex (informative): Change history

Date	Revision	Description
08/06/2023	00.00.01	Initial proposed skeleton for the TE&IV Common Topology Information Models and Interface Specification
14/06/2023	00.00.02	Updated skeleton for the TE&IV Common Information Models and Interface Specification as per comments
09/11/2023	00.00.03	Updated title for this specification and removed Stage-2 wording
11/01/2024	00.00.04	Updated the title, numbering, introduced structure for namespaces
06/03/2024	00.00.05	Merged the TE&IV Stage-2 IM CRs for March'24 train SYM.AO-2023.10.05-WG10-CR-001-TEIV NF deployment model-v07, ERI-2024.02.08-WG10-CR-0064-Adding introduction scope in TEIV IM specification-v04, ERI-2023.12.21-WG10-CR-0056-Update to TE&IV-IM specification-v01
13/03/2024	00.00.06	Editorial changes to update reference names, version as per O-RAN guidelines. Removed redundant versioning text in foreward and updated the O-RAN release versioning format in scope
13/03/2024	01.00.00	Editorial Comments, First Publication for the March 2024 Train
11/07/2024	02.00.00	Implemented the following CRs for the July'24 train: ERI.AO-2024.03.11-WG10-CR-0067-RAN-Logical-To-Cloud-Relationship-Namespace-v06, ERI-2024.04.18-WG10-CR-0071-Update to O-RAN TE&IV Network Function Deployment -v06, ERI-2024.06.10-WG10-CR-0075-Addition of UML notation and model element -v02, ERI-2024.06.10-WG10-CR-0084-adding attributes to RAN logical namespace-05, Editorial corrections.
15/07/2024	02.00.00	Addressed WG10 review comments. Used Atlassian versioning to track revisions on the captured comments.
06/11/2024	03.00	Implemented the following CRs for the November 2024 train: ERI-2024.09.18-WG10-CR-0113-TE&IV_IS_Definitions_of_ServiceOperations-v02, ERI-2024.09.18-WG10-CR-0114-TE&IV_IS_Definitions_of_ServiceOperations_for_Relationships-v01, ERI-2024.10.04-WG10-CR-0124-REL-RAN-Cloud Namespace
07/02/2025	04.00	Implemented the following CRs for the March 2025 train: ERI.AO-2025.01.15-WG10-CR-0144-TE&IV-AbstractClasses v02, ERI.AO-2025.01.28-WG10-CR-0145-RAN Abstractions-v01, ERI.AO-2025.01.27-WG10-CR-0146-REL-Cloud-RAN Abstractions-v01, ERI.AO-2025.01.27-WG10-CR-0147-TE&IV-Modelling-Guidelines v01, KDDI.AO-2025.02.19-WG10-CR-0003-TEIV-Physical v03, KDDI.AO-2025.02.19-WG10-CR-0004-TE&IV-Physical_NF_REL v02, ERI-2025.02.17-WG10-CR-0150- stage 2 to add query parameter to topology entity relationship resource -v01
01/07/2025	05.00	Implemented the following CRs for the July 2025 train: ERI.AO-2025.04.02-WG10-CR-0168- ORU-as-a-type-of-PhysicalAppliance-v02, ERI-2025.03.24-WG10-CR-0161- Classifiers-and-Decorators-v02, ERI-2025.05.20-WG10-CR-0174-Classifier-and-Decorator-Attributes-and-Service-Operations-v03, ERI-2025.06.23-WG10-CR-0184-Update OAM Arch title in TE&IV CIMI-v01, ERI-2025.06.25-WG10-CR-0187-Stage 2 Schema Operations-v02, Updated the specification for formatting corrections and editorial updates based on ODR and TS template v04

